

Distributed Resource Scheduling in Edge Computing: Problems, Solutions, and Opportunities

Yuvraj Sahni^{a,*}, Jiannong Cao^b, Lei Yang^c, Shengwei Wang^a

^aDepartment of Building Environment and Energy Engineering, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong

^bDepartment of Computing, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong

^cSchool of Software Engineering, South China University of Technology, Guangzhou, China

Abstract

Edge computing has become popular in the last decade and will advance in future to support real-time actionable analytics at the devices. One of the fundamental problems for future edge computing is to make distributed resource scheduling (DRS) decisions both at the end devices and edge devices to support requirements including autonomous computation, scalability, low-latency, etc. Several surveys in the literature on edge computing have considered some aspects related to DRS, such as challenges and solution approaches, particularly for computation offloading and data management. However, to the best of our knowledge, there is no comprehensive survey on DRS in edge computing. This paper surveys the challenging issues, motivations, and existing works for enabling DRS in edge computing. We define and identify the unique issues for DRS in edge computing compared to traditional works on parallel and distributed systems. The motivations for DRS in edge computing have been described by pointing out the benefits and emerging application scenarios. This paper also provides a taxonomy to classify the existing works from three perspectives, i.e., systems, problems, and solution approaches. Finally, we have outlined several future directions that can help researchers to advance the state-of-the-art.

Keywords: Distributed Resource Scheduling, Survey, Edge Computing, Internet of Things

1. Introduction

Edge computing has become popular in the past decade to address the needs of the emerging Internet of Things (IoT) applications such as autonomous vehicles, augmented reality, Industry IoT, etc. which have stringent requirements on latency, reliability, and scalability[1]. Edge computing pushes computation, storage, and other services closer to data sources to address concerns with cloud computing such as cost, latency, bandwidth congestion, scalability, and privacy. Furthermore, edge computing will continue to evolve to support real-time actionable analytics at the devices and support intelligence within the network. The emerging IoT applications consists of large number of devices which interact and coordinate with each other to provide distributed intelligence [2].

One of the fundamental problems for edge computing is resource scheduling which includes deciding when, where, and which resources are utilized for deploying different applications. There are different types of resources including computation, storage, network, and data, that are available at different devices. One approach to managing the

different resources is to use a centralized controller with the global knowledge of all the devices and network. It is easier to use the centralized resource scheduling approach as the controller has global knowledge and can communicate with all the devices. However, centralized resource scheduling has many issues including poor scalability, bandwidth congestion while sending all the data to a single controller, and single point of failure. Another approach is distributed resource scheduling (DRS) where control of resources is distributed among several devices that communicate with each other.

The new requirements and application scenarios of emerging IoT applications have led to the paradigm shift from centralized to distributed resource scheduling for edge computing. Distributed solution is better for resources that are distributed in space, time, or functionality [3], as is the case for edge computing. However, there are many challenges in developing a DRS solution as different devices do not have global knowledge of the whole network and, thus, need to coordinate with each other, leading to additional communication overhead. Therefore, DRS solutions need to be designed such that the communication overhead is not significant and it can support flexibility, scalability, and other requirements in edge computing.

Many existing works have studied different resource scheduling problems such as computation offloading,

*Corresponding author

Email addresses: yuvsahni@polyu.edu.hk (Yuvraj Sahni), jiannong.cao@polyu.edu.hk (Jiannong Cao), sely@scut.edu.cn (Lei Yang), shengwei.wang@polyu.edu.hk (Shengwei Wang)

Reference	Applications	Architecture Techniques	Platform Solutions	Computation Scheduling	Network Scheduling	Data/storage Scheduling	Collaborative Scheduling	Distributed Solutions
[1]	✓	✗	✗	✗	✗	✗	✗	✗
[4]	✓	✗	✗	✓	✓	✗	Partial	✗
[5]	✓	Partial	Partial	✓	✗	✗	✗	✗
[6]	✓	✗	✗	✓	✓	✓	✗	✗
[7]	✓	✓	✗	✓	✗	✗	✗	✗
[8]	✗	Partial	✗	✓	✓	✓	✗	Partial
[9]	✓	✓	Partial	✓	Partial	✓	Partial	Partial
[10]	✗	✗	✗	✓	✓	✓	✓	✗
[11]	✗	✗	Partial	✗	✗	✓	✗	✗
[12]	✗	✓	✓	✓	✓	✓	✗	Partial
[13]	✗	✗	✗	✓	✗	✓	✗	✗
[14]	✓	Partial	Partial	✓	✗	✗	✗	✗
[15]	✗	Partial	Partial	Partial	Partial	✗	✗	Partial
[16]	✗	Partial	Partial	Partial	✗	✗	✗	✗
[17]	✗	✗	✗	Partial	✗	✗	✗	✗
[18]	✗	Partial	✗	✓	✗	✗	✗	Partial
[19]	✓	✗	✗	✓	✗	✗	✗	Partial
[20]	✓	Partial	✗	Partial	Partial	Partial	✗	Partial
[21]	✓	Partial	✗	✓	✓	✗	Partial	Partial
[22]	✓	✗	✗	✓	✓	✗	Partial	Partial
[23]	Partial	Partial	Partial	✓	Partial	✓	Partial	Partial
[24]	Partial	Partial	Partial	✓	Partial	✓	Partial	Partial
[25]	✓	✗	✓	✓	✓	✓	✓	Partial
This survey	✓	✓	✓	✓	✓	✓	✓	✓

Table 1: Comparison of Existing Surveys on Edge Computing

caching, bandwidth allocation, power control, etc. Furthermore, new approaches have also been proposed for distributed control of resources in edge devices. However, these are still several opportunities to be explored in proposing DRS solutions for edge computing. Therefore, this survey investigates the existing works on DRS in edge computing to identify the challenging issues, problems studied in the literature, different solutions approaches, and future directions to advance the state-of-the-art in this area.

1.1. Related Surveys

There are several related survey papers published in literature on edge computing [4] [1] [6], different related edge computing paradigms [13], security and privacy [5] [26] [27] [21], and resource management in edge computing [12] [10] [18] [25]. A comprehensive investigation of resource management in fog/edge computing has been done in [12]. The survey in [12] gave a taxonomy to classify works related to resource management in terms of architectures, infrastructure, and algorithms. A recent survey [25] investigated resource scheduling in edge computing including distributed scheduling techniques. Another survey in [10] gave a taxonomy to classify works related to resource management in edge computing covering four aspects including resource type, objective, resource location, and resource use. A recent survey in [25] covered several aspects including architecture, methods, applications, and research directions. Some survey papers have also focused on specific problems in edge computing such as computation offloading [14] [7], caching [9], distributed data management problem [11], mobility-induced service migration [15], service placement problem [17], virtual machine management [16], etc.

There are also some survey papers about specific solution approaches such as game-theory or deep learning in edge computing. The survey in [20] investigated use of deep learning for different resource scheduling problems including adaptive caching, offloading, and edge maintenance and management. Another related survey in [19] focused specifically on machine learning-based computation offloading mechanisms. The use of machine learning-based mechanisms for reliable resource provisioning has also been investigated in [28]. A comprehensive investigation on use of economic and pricing models for resource scheduling in edge computing has been done in [8]. Another work in [29] also surveyed the use of game theory for multi-access edge computing including different use cases, types of games, and future directions.

Resource scheduling has also been investigated in related surveys for distributed systems, Cloud, 5G networks, etc. The work in [30] investigated and provided a taxonomy on resource management for grid systems. Another survey in [31] investigated techniques and solutions for improving the energy efficiency of computing and network resources in large-scale distributed systems such as grid and data centers. The survey in [32] comprehensively investigated the literature on resource sharing techniques, trends, and challenges for future communication networks including 5G. This survey [32] also provided a taxonomy for resource sharing in future communication networks. Another survey in [21] investigated integration of multi-access edge computing (MEC) with 5G technologies including non-orthogonal multiple access (NOMA), wireless power transfer (WPT), and energy harvesting (EH). DRS is an important challenge for MEC with NOMA that especially requires considering joint scheduling of computation

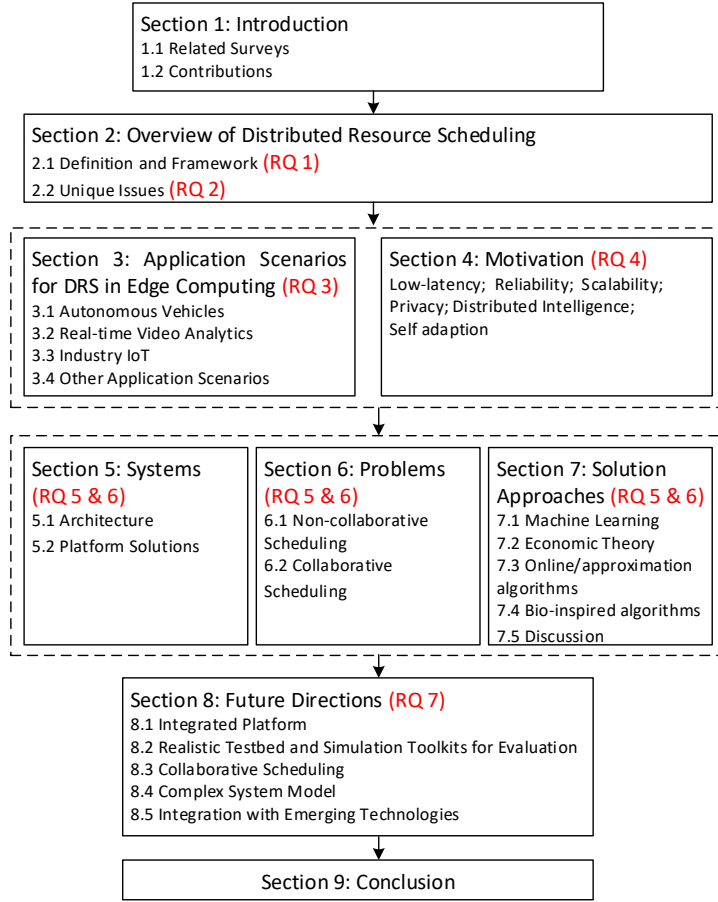


Figure 1: Paper Organization

and network resources. The work in [33] surveyed stream processing engines and mechanisms for providing resource elasticity in cloud computing.

Table 1 shows the comparison of previous survey papers with this survey paper in terms of different topics being studied in this paper. The different aspects used for comparison in Table 1 are: Applications, Architecture Techniques, Platform Solutions, Computation Scheduling, Network Scheduling, Data/Storage Scheduling, Collaborative Scheduling, and Distributed Solutions. As observed in Table 1, existing survey papers have not fully explored different types of scheduling problems especially collaborative scheduling problem. Existing survey papers have done limited investigation on platform solutions and different architecture techniques, i.e., Blockchain-based, SDN-based, and Virtualization-based, for developing distributed resource scheduling systems. Furthermore, existing surveys usually focus on centralized scheduling solutions and there is limited discussion on distributed solutions for resource scheduling in edge computing. Compared to the existing surveys on edge computing, this survey paper focuses specifically on DRS in edge computing. To the best of our knowledge, there is no existing survey paper in literature

about DRS in edge computing. Although some papers have covered part of topics such as challenges or solution approaches for DRS, particularly for computation offloading, data management, etc. However, there is a gap in literature on comprehensive review of DRS in edge computing. Recently many works have studied different types of problems for DRS in edge computing and proposed several solutions under different assumptions and constraints. The existing survey papers do not provide a systematic review of the literature on DRS in edge computing.

1.2. Contributions

This paper aims to answer several research questions related to DRS in edge computing that have not been considered in previous survey papers. To answer these questions, we have done a comprehensive investigation of related works on DRS in edge computing by searching papers through several sources including Google Scholar, IEEE Xplore, ACM digital library, and Elsevier. The research questions (RQs) investigated in this paper are:

- RQ 1: What is DRS in edge computing?

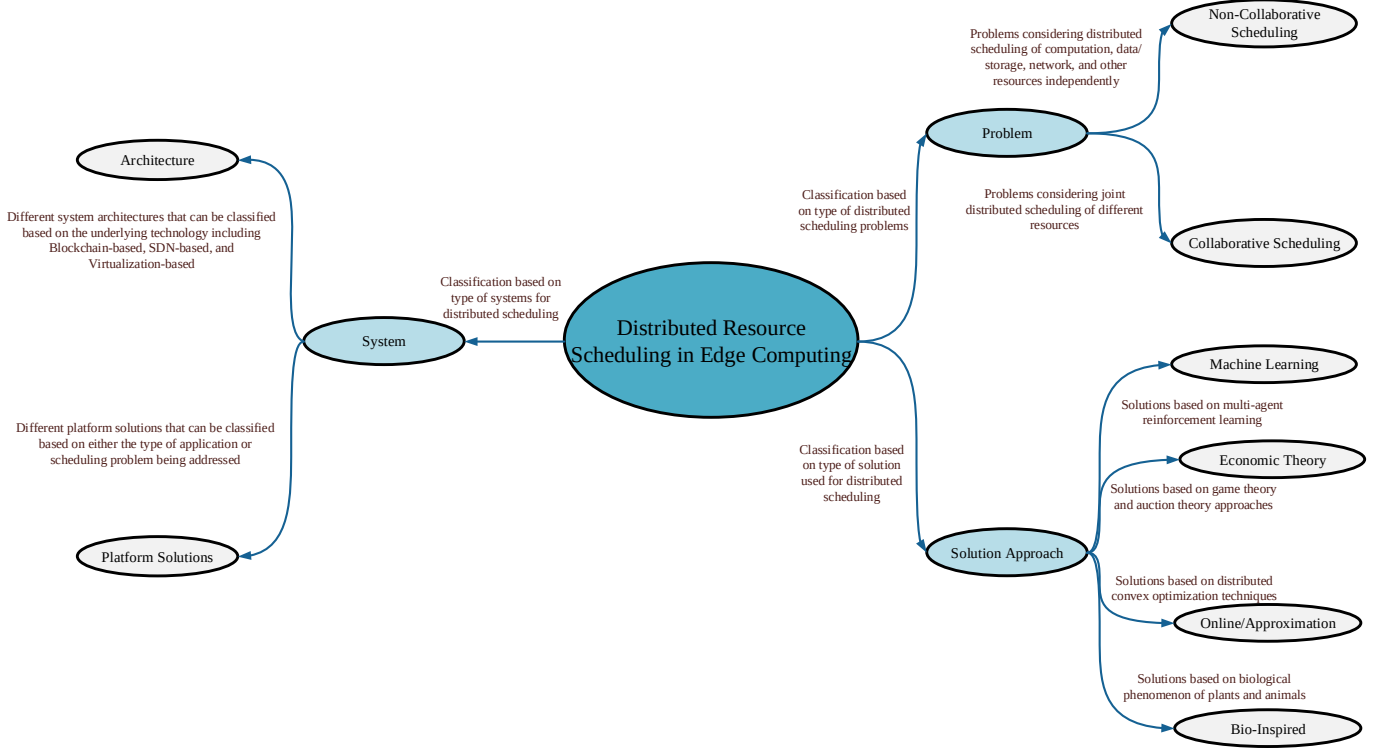


Figure 2: Taxonomy for Classification of Existing works for Distributed Resource scheduling in Edge Computing based on System, Problem, and Solution Approach

- RQ 2: What are new unique issues for DRS in edge computing compared to traditional parallel and distributed systems?
- RQ 3: Which application scenarios are enabled by leveraging DRS in edge computing?
- RQ 4: What are benefits of enabling DRS in edge computing?
- RQ 5: What is the taxonomy to classify different existing works on DRS in edge computing?
- RQ 6: What are different systems, problems, and solutions approaches for DRS in edge computing?
- RQ 7: What are future research directions for DRS in edge computing?

By answering the above research questions, this paper will enable researchers to have an in-depth understanding of DRS in edge computing. The overall organization including different sections and research questions answered in each section is shown in Fig 1. The main contributions of this paper lies in covering the following four aspects related to DRS in edge computing.

1. We define DRS in edge computing and describe a reference framework for edge computing that gives a high-level view of edge computing architecture. The

framework helps in highlighting the central role of resource scheduling in the edge computing architecture to provide services for different applications. We also identify several unique issues for DRS in edge computing by comparing resource scheduling characteristics in traditional parallel and distributed systems with resource scheduling in edge computing.

2. We describe several scenarios in emerging application domains that can benefit from DRS. We also point out the motivations for enabling DRS in edge computing. The use of DRS in edge computing leads to several benefits such as low-latency, reliability, scalability, privacy, etc.
3. We provide a taxonomy to classify existing works from three perspectives: systems, problems, and solution approaches, as shown in Fig 2. The details and further sub-categories within each category are explained in this survey. For example, system category includes architecture and platform solution. We also provide our observations based on the investigation and summarise existing works by classifying them based on the type of problem and solution approaches.
4. We also outline several future directions including developing integrated platforms, realistic testbeds and simulation toolkits, collaborative scheduling so-

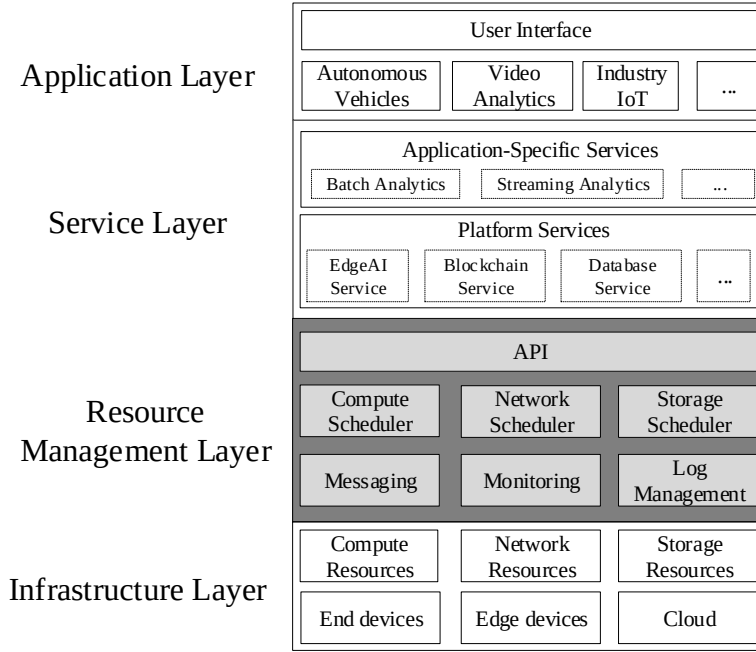


Figure 3: Reference Framework of Edge Computing

lutions, etc. that provide opportunities for researchers to improve the state-of-the-art for DRS in edge computing.

The rest of the paper is organized as follows. Section 2 describes the definition and unique issues of DRS in edge computing compared to traditional works in parallel and distributed systems. Section 3 describes application scenarios for DRS in edge computing. Section 4 points out the motivations for using DRS in edge computing. Section 5 describes the different systems for DRS in edge computing. Section 6 mentions the different types of problems studied for DRS in edge computing. Section 7 discusses different solution approaches developed to enable DRS in edge computing. Section 8 outlines the future directions that provide opportunity for researchers. Finally, the conclusion is given in Section 9.

2. Overview of Distributed Resource Scheduling

This section first defines DRS and introduces the reference framework of edge computing. Then, it describes the unique challenging issues to consider for enabling DRS in edge computing.

2.1. Definition and Framework

DRS in edge computing refers to distributed scheduling of different resources available in edge computing environments to optimize system and/or application-specific objective. The resources implies any reusable entity that is

utilized to fulfill an application or resource request [30]. Resources in edge computing can include computation resources (CPU/GPU, memory, etc.), storage resources (disk, flash, etc.), network resources (bandwidth, wireless channels, etc.), or even services that are provided by composing multiple resources. Scheduling implies decisions on when and which resources are utilized while considering the constraints. Edge computing can be used for different purposes such as providing computation resources close to data resources, sharing data among different devices/entities, or providing other services. Therefore, the scope of resource scheduling depends on the type of resources involved as shown by different types of problems described in Section 6. The main keyword in DRS is distributed that refers to distributed control of resources. The work in [12] distinguishes the control of resources in terms of centralized and distributed. For centralized control, a single controller or central algorithm is used to manage and allocate the resources. In contrast, distributed control corresponds to use of multiple controllers or distributed algorithm for managing and allocating the resources in edge computing environments.

We provide a framework for edge computing to help understand the context and aspects to be considered for DRS in edge computing. The framework, shown in Fig 3, consists of four layers including infrastructure layer, resource management layer, service layer, and application layer. As shown in Fig. 3, there are different types of devices in an edge computing infrastructure. The different resources distributed across the infrastructure can be

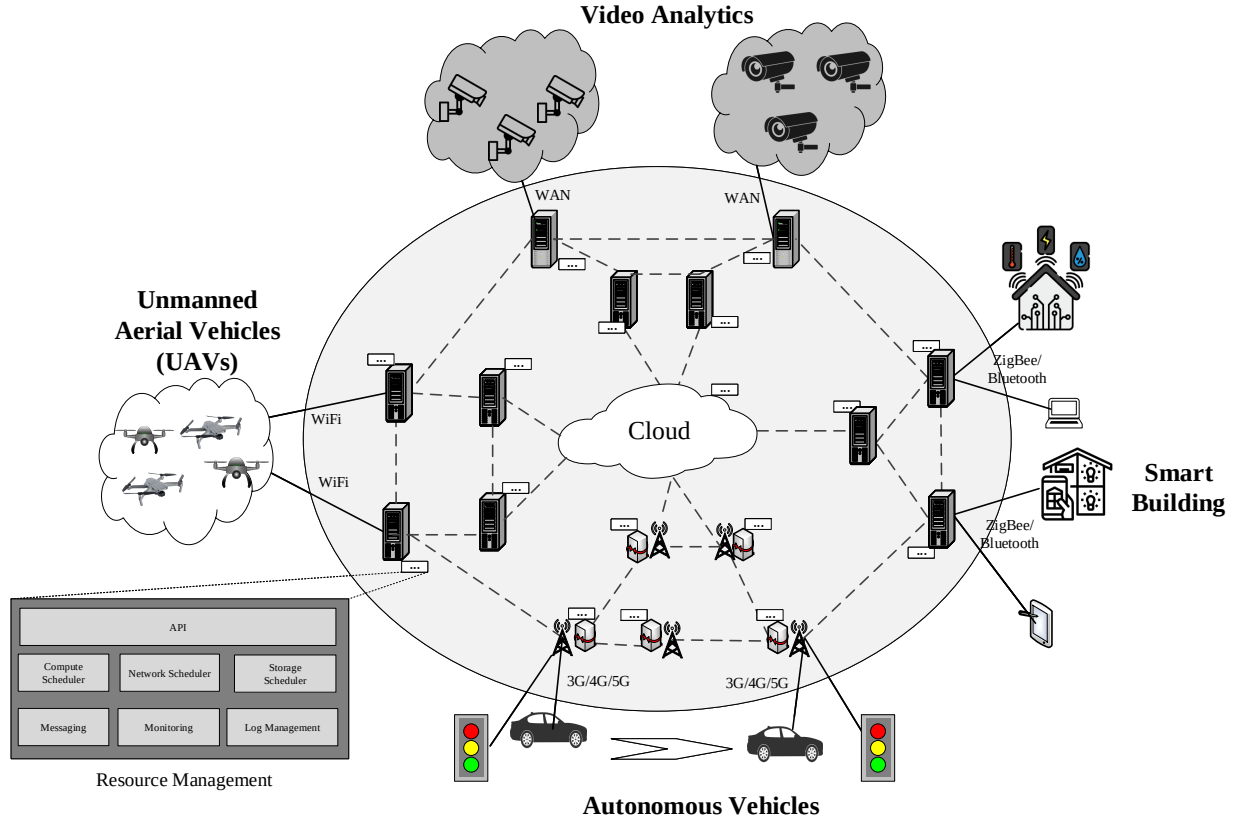


Figure 4: Distributed Resource scheduling in Edge Computing

abstracted as compute resource, network and storage resource. One of the main issue to be addressed for edge computing is managing the distributed resources to deploy applications and platform services. As shown by the highlighted resource management layer in Fig. 3, there are several components required to monitor and schedule the different resources. The different application and platform services are developed and deployed on the edge computing infrastructure by leveraging the underlying resource management layer. Finally, different edge computing applications can be deployed and executed using the different services. The details of each layer with the framework is described as follows:

1. **Infrastructure Layer:** Infrastructure layer consists of different types of devices and resources available in the edge computing system. The devices can be classified as an end device, edge device, or a cloud server and the resources can be abstracted as a compute resource, storage resource, network resource, etc.
2. **Resource Management Layer:** Resource management layer includes different components used for monitoring and scheduling the resources. The different resources are managed by the several compo-

nents each responsible for a specific functionality. In case of centralized resource scheduling, the scheduling decisions are made at a centralized controller. For example, centralized controller in cloud computing paradigm monitors and makes scheduling decisions for all the resources. On the other hand, DRS implies that different devices can be autonomous and make the scheduling decision for resources independently. Fig 4 shows an example scenario of DRS in edge computing where there are different edge devices inside the circle connected in a mesh-network architecture. Each edge device has all the components in the resource management layer and can act independently to make scheduling decision. The details of each component are as follows:

- **Messaging:** responsible for communication with other devices
- **Monitoring:** responsible for monitoring the availability and consumption of resources
- **Log Management:** responsible for logging the scheduling decision and resource consumption
- **Scheduler (Compute, Network, Storage, etc.):** responsible for scheduling different resources. For example, Compute Scheduler decides when

and where each task is executed, Network Scheduler decides on start time, bandwidth consumed, routing path, etc. for network flows, and Storage Scheduler decides which device is used for caching particular type of data.

- **API:** responsible for interfacing with the upper layers in the framework including user interface. For example, upper layers can use resource management layer API to make computation scheduling decisions for application tasks.
3. **Service Layer:** Service layer consists of different services provided by edge computing system. These services can be abstracted as either platform-specific services or application-specific services. Platform service includes Edge AI service, Blockchain service, Database service, and other services that allows users to develop and deploy applications without focusing on the complexities of managing the different resources within the infrastructure. Application service includes batch analytics, streaming analytics, and other services that are directly utilized by users to build application.
 4. **Application Layer:** Application layer includes different types of edge computing applications such as autonomous vehicles, video analytics, Industry IoT, etc. that can be built and deployed on the edge computing infrastructure. Application layer also includes user-interface to connect with the users.

2.2. Unique Issues

There are many issues to be considered for DRS in edge computing compared to centralized resource scheduling. These issues include the availability of partial information at each device, limited access of resources at different devices, synchronization of control data among different devices, the convergence of the distributed solution, lack of trust among devices from different stakeholders, etc. DRS has been well-studied previously for traditional parallel and distributed systems [30]. However, there are many unique issues in DRS in edge computing compared to traditional parallel and distributed systems. Table 2 shows the comparison of the characteristics of resource scheduling in parallel and distributed systems with resource scheduling in edge computing. The comparison shows several differences in terms of different criteria, including computation resources, network, data, etc. These differences affect the metrics, including performance, scalability, reliability, security, and privacy. The unique issues for DRS in edge computing can be summarized as follows.

1. **Heterogeneity:** The devices in the network are usually resource-constraint and heterogeneous in terms of computation capability, storage, and other resources. There can be multiple simultaneous applications running leading to dynamics in resource

availability on devices. These devices are also power-hungry and prone to failures. DRS solution proposed for edge computing should be adaptive and consider the vagaries of the available resources in edge computing.

2. **Network dynamics and wireless interference:** The network in edge computing is also heterogeneous in terms of communication protocols, i.e., it can have both wired and wireless communication protocols. The wireless network, especially, is unstable and has limited bandwidth. There can be multiple simultaneous data transmissions in the network leading to wireless interference. DRS solutions also need to consider wireless interference, which is usually ignored for scheduling solutions in traditional parallel and distributed systems. Another characteristic feature of the network is spatial and temporal dynamics and an open environment with mobile and new participants leading to additional complexity. Therefore, DRS solution needs to be adaptive and consider different resource metrics related to the network, including bandwidth allocated, selected channel, routing paths, etc.
3. **Device access and resource availability:** The network in edge computing is heterogeneous, consisting of different types of devices, including end devices, edge devices, and cloud [2]. Furthermore, there can be different hierarchies in edge devices as defined in fog computing. The different devices have heterogeneous accessibility and resource availability. For example, end devices can access the nearby edge device, whereas edge devices can access resources from other edge devices and clouds. The heterogeneity in device access and resource availability makes it challenging as DRS solutions deployed on end devices or edge devices should consider the type of device and resources available to manage at the device.
4. **Data generation and storage:** Application tasks and data are tightly coupled in edge computing as the tasks are generated and executed by devices near data sources [34] [35]. A characteristic feature for edge computing is distributed data sources, i.e., data required for a task is usually generated from multiple IoT devices which can be connected to different nearby edge devices [36]. For example, many machine learning applications use multi-modal data from different sources to improve accuracy. In these cases, the data required for different application tasks can be stored or generated at different devices. Therefore, DRS solutions should jointly consider task and data management, i.e., task scheduling decisions should be made considering the data placement and storage. Most IoT applications are also dependent on the quality and source of data required for task execution. There can be redundant

Criteria	Resource Scheduling in Parallel and Distributed Systems	Resource Scheduling in Edge Computing	Affected Metric
Computation resources	• Powerful servers and processors; • Static and dedicated environment with fixed nodes	• Resource-constrained lightweight devices; • Dynamic and open environment with mobile and new participants	• Performance; • Scalability • Reliability
Network	• Flat and homogeneous network; • Stable network and guaranteed-bandwidth; • Dedicated wired network for communication;	• Hierarchical and heterogeneous network; • Unstable network and limited bandwidth; • Heterogeneous wired and wireless network with interference for communication;	• Performance; • Scalability; • Reliability
Data	• Input data stored in nodes; • Single source for input data	• Data generated in real-time; • Multiple data sources	• Performance; • Scalability
Applications	• Top-down released and executed; • Compute-intensive and location-unaware; • Fixed decomposition before runtime;	• Bottom-up released and executed • Data-intensive and location-aware; • Changeable decomposition in runtime	• Performance; • Scalability
Management	• Decoupling of resources (independent management of data caching, network scheduling, task scheduling, etc.)	• Tightly coupling of resources (joint management of data caching, network scheduling task scheduling, etc.)	• Performance
Environment	• Trustable environment (no privacy issue); • Usually single administrator/owner of resources	• Untrusted environment (data are privacy-sensitive); • Multiple owners/administrators of resources	• Security and Privacy
Target	• Parallelism and resource utilization	• Low latency and energy consumption	• Performance

Table 2: Characteristics of Resource Scheduling in Edge Computing

sources for the same type of data, and the quality of data can be adjusted. Therefore, the scheduling decisions should consider the different data properties, including quality, source, and data type required for a task.

5. Devices belonging to different stakeholders:

The resources in edge computing can be under the ownership of different stakeholders leading to the requirement of novel pricing models and incentive mechanisms that promote the sharing of resources. DRS solutions should be designed considering incentive mechanisms for resource sharing. One of the main issues is trust and governance concern due to different stakeholders in edge computing [37]. Many recent solutions have also adopted Blockchain to address some of the concerns associated with different stakeholders. In the case of blockchain-enabled edge computing systems, resource scheduling requires jointly considering blockchain-specific parameters, and new consensus mechanisms should also be designed.

6. Application adaptability: The applications do not have a fixed decomposition and can also be adapted at run-time, i.e., application-related parameters can be adapted based on the resource availability to adjust the quality of service. For example, in deep learning applications, we can compress/quantize the deep learning model depending on the network condition, accuracy requirement, etc. Another example is video caching and delivery, where we can change video quality by compressing at different resolutions. Therefore, DRS solutions need to be designed to optimize the network and system resources and application-related parameters jointly. The joint optimization is challenging as the problems need to be modelled and solved considering the complex dependency among the parameters.

7. Security and privacy Issues: Many traditional security and privacy mechanisms are not directly applicable in edge computing due to the use of resource-constrained edge devices and dynamic wireless networks. There are several security and privacy concerns in edge computing as studied in existing works [38] [26] [27]. The design of DRS solutions for edge computing requires sharing data and resources among heterogeneous devices in edge computing using wireless networks, which poses more vulnerabilities. Many issues such as identity management, authorization, authentication, etc., need to be considered in developing DRS solutions for edge computing.

3. Application Scenarios for DRS in Edge Computing

There are several application scenarios that can benefit from DRS solutions in edge computing. This section points out some of the emerging application scenarios in edge computing.

3.1. Autonomous Vehicles

Autonomous vehicles is one of the major applications for edge computing. Several use cases have been identified for autonomous vehicles in existing works [39], [40]. The use cases identified in [40] include convoy driving, cooperative lane change, and cooperative intersection management. Convoy formation is vehicles grouped in a stable formation to form a platoon. The vehicles in convoy need to share information frequently to maintain the convoy formation. The work in [41] highlights the use of a distributed control algorithm for multi-lane convoy formation that has better scalability. Cooperative lane change is another use case that can benefit from DRS as vehicles need to collaborate by sharing their planned trajectories with each other. Cooperative intersection management involves vehicles crossing an intersection without the use of a traffic light. The work in [42] proposed a distributed solution for vehicles to autonomously decide their velocity in order to utilize the traffic intersections efficiently. Congestion detection and avoidance is another use case presented in [2]. Congestion detection and avoidance involve sharing data among vehicles to detect the traffic and re-route the vehicles accordingly. A fully distributed algorithm has been proposed for congestion detection in [43]. Another work [44] proposed a distributed algorithm to re-route traffic for congestion avoidance.

There are many other use cases in autonomous vehicles, such as intelligent parking, accident prevention, collision warning, etc. These different use cases involve addressing challenges such as cooperative decision, and cooperative sensing [39]. Many recent works have proposed solutions to address the DRS in autonomous vehicles. For example, the work in [45] proposed a software-defined collaborative vehicle edge computing framework to manage the network resources collaboratively. Distributed platforms have also been proposed in works such as [46], [47], etc., to enable autonomous vehicles.

Need for DRS solutions: The different use cases identified for Autonomous Vehicles needs to support many requirements, including high and constrained mobility, real-time processing, reliability, spatial and temporal dependence of data, etc. [2] [40]. It is challenging to support these requirements using centralized resource scheduling. A large amount of data is generated from various infrastructure sensors and onboard sensors to be used by different vehicles travelling at high velocity. Autonomous vehicles need to analyze and share data to make decisions related to object detection and tracking, action prediction, path planning, and obstacle avoidance, etc. The data

needs to be analyzed locally at the devices as each data is associated with particular age and value. Furthermore, the data is only valid for a short time and distance. For example, the vehicle deciding on obstacle avoidance cannot use data from a different road segment one hour ago. A centralized controller cannot be used to make the resource scheduling decision for vehicles as it would lead to communication bottleneck and significant delay if data needs to be sent from all the vehicles. Furthermore, the vehicles usually have different owners who may not be willing to share the data to a centralized controller due to privacy concerns [2]. Reliability and safety are also crucial for autonomous vehicles as any issue in resource scheduling and sharing could lead to accidents and loss of life in severe cases. These different concerns cannot be addressed by using centralized resource scheduling and thus, give a strong case of using DRS in edge computing.

3.2. Real-time Video Analytics

Real-time video analytics has been referred to as a killer application for edge computing [48]. The work in [49] describes use cases of real-time video analytics for public safety such as police applications, transportation applications and firefighting applications. These use cases focus on situation awareness, alerting, and detecting objects that could be criminal or missing. For example, in police applications, real-time video analytics can be specifically used to detect suspicious criminals in real-time in a surveillance zone. Police personnel can also use real-time analytics to find missing children using city-wide video surveillance. Firefighting applications can use real-time video analytics to monitor forest fire or emergency building evacuation during a fire [2]. These applications can also use multiple unmanned aerial vehicles (UAVs) to analyze the video from multiple cameras collaboratively.

Another use case of real-time video analytics has been given in [50] for achieving the goal of Vision Zero. Vision Zero is an international initiative to prevent road accidents by following a proactive approach to identify risks. Real-time video analytics is a core component to achieve the goal of Vision Zero as highlighted in [50]. A practical implementation on the use case of real-time video analytics is shown in [50] where Microsoft, City of Bellevue, WA, USA, and the University of Washington (UW) developed a system to use the city's existing traffic cameras to identify near-collision events in real-time. The system generates real-time reports on traffic, including the number, type, and direction of vehicles. The systems can also detect and follow the trajectory of vehicles and give information of collision-prone areas.

Real-time video analytics has use cases in many other application domains too such as smart home, smart healthcare, Industry IoT, etc. Several solutions have been proposed in the literature to perform real-time video analytics. The work in [51] proposed a distributed architecture for object detection in edge computing. Another work [52] proposed a distributed framework for video surveillance.

EdgeEye [53] is another framework proposed for real-time video analytics. The work in [54] proposed a solution, Convince, for collaborative video analytics.

Need for DRS solutions: Real-time Video Analytics has requirements to support real-time processing for critical applications such as AR/VR, vehicular network, etc. Video analytics involves streaming data with high bandwidth and computation requirements. These requirements make it challenging to use a centralized controller for processing. Hence, the work in [48] highlights the use of distributed cloud and edge architecture for large-scale real-time video analytics. Large-scale video analytics also poses overhead challenges in terms of energy consumption, processing throughput, etc., while using deep learning algorithms for analytics. The overhead can be reduced by enabling collaboration among video cameras to perform cross-camera video analytics [55]. Collaboration is an essential requirement to reduce overhead and share results among devices for providing better services. Privacy is also another concern as different stakeholders can own the video cameras. Therefore, DRS is vital in achieving the goals of different use cases due to the requirements on large-scale, real-time processing, and collaboration among cameras to share data.

3.3. Industry IoT

Industry IoT is another primary application domain for edge computing. Industry IoT systems can be considered as a collection of autonomous, interconnected components that collaborate to appear as a single coherent system [56]. Industry IoT covers several different applications under different industries such as Smart Manufacturing, Smart Logistics, Smart Construction, etc. The work in [57] provides several use cases for Smart Manufacturing. Smart manufacturing involves using connected smart devices to improve the efficiency of operations, management, and maintenance in manufacturing plants. Production optimization is a critical use case where edge computing can be used to track and compare the performance of different components in the manufacturing plant to determine performance bottlenecks. Edge computing can also be used for process optimization where parameters such as temperature, humidity, pressure etc., are optimized for different processes.

Another use case within smart manufacturing is prognostics and health management (PHM) [58], which includes several aspects such as performance detection, fault detection, fault isolation, fault prediction, enhanced diagnosis, health management, component life tracking, etc. One example under PHM is predictive maintenance, where edge devices can collaborate to predict and prevent failures in real-time by dynamically configuring functionalities of different processes.

Smart Warehousing and logistics is another significant use case within the Industry IoT application domain. Smart logistics involves full automation of different operations in a warehouse such as storage, packaging, sorting,

stacking, delivery, recording, management, etc. [59] [58]. Besides automation, other requirements such as adaptivity and low latency are crucial as new orders arrive each day. These operations often require the use of a multi-robot system where different robots collaborate to complete the task [59]. One example use case is described in [58] for classification and sorting process where different robots and edge servers collaborate to first recognize the good at the terminal based on RFID, plan the path from picking location to terminal, transport the good, and finally verify it again at the terminal. There are other use cases too in a smart warehouse such as monitoring the transport of goods within a warehouse or transport from one warehouse to another.

Several other use cases can also be found in smart agriculture, smart construction, smart grid, etc. Many solutions have been proposed in the literature to enable these Industry IoT use cases. For example, the work in [60] described the iRobot Factory system and compared it with traditional factory systems. Another work in [61] proposed a collaborative tracking system for manufacturing plants.

Need for DRS solutions: The different Industry IoT applications have some common essential requirements, including high scalability, real-time processing, automatic control, dependability, flexibility, collaboration, security and privacy, etc. [62] [58]. A manufacturing plant can include several production types of equipment and processes running, which need to collaborate for optimized operation and management. Smart Manufacturing is characterized by having large-scale production, personalized and agile product development, and achieving zero inventory [57]. These use cases involve a large number of devices collaborating autonomously. Such requirements pose several challenging issues that are difficult to be addressed using a centralized resource scheduling solution.

3.4. Other Applications Scenarios

Mixed reality is another emerging use case for DRS in edge computing. Mixed reality has become popular recently due to different gaming and entertainment scenarios [63]. Furthermore, mixed reality has use cases beyond gaming or entertainment such as collaborating for a business meeting using holographic collaboration [64]. Holographic collaboration is an emerging scenario where mixed reality is used for social interaction, which can be for a business meeting, or a remote discussion. Mixed reality can also be used in smart healthcare by medical professionals to perform virtual surgery [65]. These kind of scenarios have stringent requirements on low latency and reliability, which can be addressed by DRS.

There are many use cases for DRS in edge computing involving UAVs and Multi-robots systems (MRS). Both UAVs and MRS include collaboration among different devices interacting autonomously. For example, UAVs can be used in emergency rescue and search operations [66]. Another use case using UAVs is an air-ground integrated network connecting both UAVs with the ground vehicles

and roadside infrastructures [67]. MRS can also be considered another application domain for edge computing where each robot can be modelled as an edge node. MRS can be used in several application scenarios in healthcare, warehouse, emergency rescue, etc.[68].

4. Motivations

This section points out the motivations for using DRS in edge computing. The use of DRS in edge computing provides several benefits including low-latency, better fault tolerance, higher scalability, better privacy preservation, and enabling distributed intelligence. The various benefits also correspond to characteristic requirements of applications that can benefit from DRS.

1. **Low-latency:** In general, the edge computing paradigm reduces the latency by pushing the computation from the centralized cloud server to edge devices closer to data sources. However, the different edge devices may not cooperate and could use a centralized controller such as an SDN controller to make resource scheduling decisions which leads to network congestion and high latency. DRS helps in reducing the network load and link congestion by avoiding sending the data to a centralized decision controller. The reduction in network congestion leads to direct improvement in system response time. DRS also enables devices to collaborate to make faster intelligent decisions either locally or by sending data to a nearby controller.
2. **Reliability:** Reliability is a significant concern while managing resources in edge computing as the network can be unstable and dynamic. Both the devices and network links are prone to failures. Centralized resource scheduling is not suitable as a centralized controller has a single point of failure; it can crash and fail or be attacked by an adversary leading to security issues. DRS solves this problem by providing redundancy using multiple controllers to support the high reliability and availability requirements for emerging applications such as autonomous vehicles, Industry IoT, etc. The system can continue to function by being managed by another secondary controller in case of failure of one resource controller.
3. **Scalability:** The development of new emerging application scenarios and better technology adoption has led to a huge proliferation in the number of devices and data in edge computing. These new applications such as AR/VR, smart city, etc. require scalability in terms of both devices and services. It is challenging to support scalability using a centralized controller as sending the data from all the devices leads to network bandwidth congestion and inefficient performance in terms of systems response time.

Even increasing the network bandwidth capacity is also not sustainable as the emerging applications involve high-bandwidth multimedia data. DRS can help support scalability by allowing multiple controllers to manage the resources, reducing network congestion. DRS in edge computing enables the creation of hierarchical and distributed edge computing systems that can scale the network in terms of both the number of devices, services, and geographical area.

4. **Privacy:** As mentioned earlier, resource scheduling in edge computing involves an untrustable environment that is data and privacy-sensitive. The data and devices can be owned by different stakeholders who may be unwilling to share their resources due to privacy concerns. Centralized resource scheduling is not suitable in such a scenario as it requires the data to be sent to a centralized controller. Furthermore, sending the data through a wireless network in edge computing makes it more prone to security attacks and privacy leakage. DRS address the privacy concern by allowing the different devices to make their decisions locally and share the results instead of raw data.
5. **Distributed Intelligence:** Distributed intelligence implies interaction among different devices and stakeholders that collaborate to satisfy individual and/or global goals. Edge computing involves different stakeholders that need to collaborate to share resources and support distributed intelligence. Centralized resource scheduling cannot enable distributed intelligence as all the scheduling decisions are made at a centralized controller. DRS enables intelligence at the devices by allowing different devices to make their decisions independently and enable collaboration among them by sharing the resources.
6. **Self-adaption and configuration:** Emerging edge computing applications such as autonomous vehicles, unmanned aerial vehicles (UAVs), etc., require the network and devices to be flexible and adapt according to the network conditions. As the edge computing networks can be unstable and dynamic, supporting self-adaption and configuration is an important requirement for these applications. However, it is challenging to support self-adaption for devices in an edge computing system with centralized resource scheduling as the devices cannot make their resource scheduling decision independently. DRS allows the devices to be autonomous and self-adapt in a dynamic network by distributing the resource scheduling decisions to different devices.

5. Systems

This section covers the existing works related to systems for enabling DRS in edge computing. We have further classified the related work in terms of platform solutions and architecture.

5.1. Architecture

Different types of system architecture have been proposed in the literature for enabling DRS in edge computing. In this survey, we have classified these architectures in terms of the underlying technology being used, i.e. Blockchain-based, SDN-based, and Virtualization-based. In addition, there are some works that have also combined these technologies to propose an architecture for the system.

5.1.1. Blockchain-based

Blockchain benefits edge computing by enabling collaboration and secure sharing of resources among trustless stakeholders [69]. [Blockchain-based systems specially address unique issues emerging due to devices belonging to different stakeholders, and security and privacy issues.](#) A blockchain-based framework is proposed in [70] for secure task sharing among edge servers in a mobile edge computing (MEC) environment. This work [70] also investigates the use of blockchain to provide incentives for collaboration. Another work in [71] has proposed BCEdge, a blockchain-based DRS scheme in D2D-assisted MEC. A Blockchain-based solution has been proposed in [72] to enable privacy of network topology in MEC by implementing multi-domain collaborative routing consensus mechanism.

5.1.2. SDN-based

Several SDN-based edge computing architecture have been proposed in literature where multiple SDN controllers collaborate to support DRS in edge computing. SDN helps in better resource management by separating the control functionality from the data plane. [In terms of unique issues for DRS, SDN-based systems address the issues emerging from device heterogeneity and network dynamics and wireless interference.](#) For example, the work in [73] proposed SoftEdgeNet, an SDN-based distributed layered network architecture considering several design principles including heterogeneity, real-time analytics, fault tolerance, scalability, etc. SoftEdgeNet also uses blockchain technology to meet the design principles. An SDN-based architecture has also been proposed for vehicular networks in [74] to support low-latency and high-reliability communication. Another work in [75] also proposed SDN-based resource management autonomous vehicular networks.

5.1.3. Virtualization-based

Containers and other virtualization technologies provide an abstraction to enable collaboration among heterogeneous devices. Other benefits include isolation to

enable the deployment of multiple simultaneous applications. [Virtualization-based systems specifically address the unique issues emerging from device heterogeneity in edge computing environments.](#) The work in [76] considers the use of different lightweight virtualization technologies such as Docker and Unikernel for edge computing. A distributed and lightweight virtualization model, LW-CoEdge, has been proposed in [77] to enable collaboration and address other requirements for edge computing environments. This work [77] also proposes heuristic algorithms to enable DRS and data sharing among peer-to-peer edge devices. Another work in [47] proposed a container-based offloading framework for autonomous vehicles application scenario. The proposed framework includes offloading decision module, offloading scheduler module, and an edge offloading middleware [47].

5.2. Platform Solutions

Platform solutions for DRS in edge computing provide an execution environment to support application services. The existing works on platform solutions in literature are of two types:

5.2.1. Application-specific

These platform solutions are designed considering the specific application requirements such as heterogeneity, real-time processing, mobility, etc. These solutions often include an abstraction layer to manage the heterogeneous resources. The application-specific platform solutions consider application-specific protocols and techniques, such as industry communication protocols for Industry IoT applications. [Application-based systems are designed to consider unique issues emerging from application adaptability.](#) Two popular application scenarios that have been explored in literature are autonomous vehicles and Industry IoT.

1) Autonomous Vehicles: An edge computing-based Open Vehicular Data Analytics Platform (OpenVDAP) has been proposed in [46] to enable collaboration among neighbouring vehicles, road-side servers, and cloud servers for dynamic offloading of computation tasks. A dynamic offloading and scheduling algorithm is included to make the offloading decision based on the service's status, computation overhead, and the optimal offloading destination while considering the latency and bandwidth consumption requirements. OpenVDAP consists of several components, including on-board heterogeneous computing/communication unit (VCU), a vehicle operating system (EdgeOSv) to support secure and privacy-aware running environment for resource management, a driving data integrator (DDI) to collect data from different sources, and a library to build applications using the platform. [OpenVDAP considers several unique issues described in Section 2.2 including heterogeneity, network dynamics and wireless interference, device access and resource availability, data generation and storage, and security and privacy issues.](#)

Another work in [78] designed and implemented a low-power edge computing system, LoPECS, to support real-time services for autonomous vehicles. The work in [79] also surveyed computing systems for autonomous driving.

2) Industry IoT: The work in [80] implemented a prototype platform based on the proposed architecture of edge computing for IoT-based manufacturing. The architecture uses edge computing at different levels, including individual stages such as system integration, interconnection, information fusion, or the entire life cycle of the manufacturing process. The platform utilizes Open Platform Communications United Architecture (OPC UA) and Data Distributed Service (DDS) for enabling information interaction between manufacturing resources and different control systems. Furthermore, a multi-agent system has been used to support autonomous negotiation capability in manufacturing plants. [The platform proposed in this work considers several unique issues of DRS including heterogeneity, network dynamics and wireless interference, device access and resource availability, data generation and storage, devices belonging to different stakeholders, and security and privacy.](#) Another work in [81] also proposed a distributed platform for manufacturing plants based on edge computing and OPC UA.

5.2.2. Problem-specific

These platform solutions are designed to manage a particular resource such as computation, data, network etc. Most existing works have proposed DRS platform solutions to support computation management and data management.

1) Computation Management: The work in [82] designed a platform, LAVEA, to support computation offloading between clients and edge nodes, and collaborate with nearby edge nodes to support low-latency video processing. LAVEA includes three main edge computing services, i.e. profiler, monitoring, and offloading. [LAVEA is based on server-less architecture and addresses the unique issue of heterogeneity among devices.](#) Another work in [52] also proposed a platform named SIAT for distributed video analytics. SIAT uses several state-of-the-art distributed computing technologies, including Apache Spark, Hadoop Distributed File System (HDFS), Apache Kafka, etc., to support analytics of both real-time video streams and batch video analytics. [SIAT addresses the unique issues of application adaptability by providing distributed video processing library to support large-scale video analytics.](#) Due to the rising popularity of AI, the work in [83] proposed Open Framework for Edge Intelligence (OpenEI), a framework to develop platform solutions to enable devices having different computation capabilities with edge intelligence.

2) Data Management: Firework is a data-sharing platform proposed in [84] to enable collaboration among multiple stakeholders. Firework enables privacy-preserving big data sharing and processing by fusing geographically distributed data and exposing it to the end users using virtual

shared data views. Another example of a data management platform is DataFog, proposed in [85] for the geo-distributed and heterogeneous edge computing environment. DataFog makes several design decisions, including locality-aware distributed indexing, replication policy, and location-aware load balancing to support spatio-temporal range queries in applications. The work in [86] proposed Griffin for effective sharing of data and state among multiple edge devices considering challenges such as distributed data sources, data storage abstraction and consistency, heterogeneity, network dynamics, etc. [These different data management platforms enabling data sharing by addressing the unique issue in DRM of devices belonging to different stakeholders.](#)

6. Problems

This survey classifies the different resource scheduling problems into two categories, i.e. non-collaborative and collaborative scheduling problems, depending on the number of resources being managed.

6.1. Non-collaborative scheduling problems

The different problems in this category can be classified based on the type of resource being managed. The four categories are computation management, data management, network management, and resource allocation.

6.1.1. Computation scheduling

This includes problems related to the scheduling of computation resources available at the different types of devices, including edge devices, MEC servers, IoT devices, etc. Some of the main problems studied in this category are:

- 1) Task allocation/scheduling: Edge computing system can consist of multiple IoT devices, edge devices, and the cloud. Each edge device can further consist of multiple core processors where multiple tasks can be executed in parallel. Task allocation/scheduling problem aims to allocate the available computation resources to the tasks to determine when and where each task is executed while considering the environment constraints. Task allocation problems usually consider devices where the task should be executed, whereas task scheduling problems also consider the order of computation tasks to determine when the task should be executed. Existing works have also considered whether the different edge devices can be cooperative, i.e. computation resources can belong to a single owner, or competitive, i.e. computation resources can belong to multiple owners. This problem has been studied for various objective metrics, including minimizing delay, energy consumption, load balancing, system cost, etc [14] [87].

- 3) Computation offloading: Computation offloading can be considered to be a specific case of task allocation/scheduling problem. Computation offloading includes migrating the computation tasks generated at IoT devices

or edge devices, either partially or fully, to a selected edge device. The problem consists of selecting the edge device, i.e., where the computation task would be executed. In the case of partial offloading, the problem also includes determining the partitioning ratio of the tasks [88]. Several works have also considered determining the start time of tasks, i.e., when the computation tasks are executed, as part of the computation offloading decision. Existing works have considered the computation offloading problem for different system and application models such as independent tasks, dependent tasks, single server, multi-server, single-hop, multi-hop, etc [14] [7].

- 3) User allocation: User allocation problem includes deciding to allocate the amount of computing resources at the edge devices/server to application users. This problem usually aims to maximize the number of allocated users from the application vendor's perspective. Existing works have also considered minimizing the system cost and achieve load balancing among edge devices/servers to better utilize the resources [89] [89] [90].

- 4) Server activation: Service activation includes deciding whether the server can accept (active mode) or not accept (inactive mode) the computation tasks. This problem is considered while offloading computation tasks for cases where the edge server/device can choose to accept the computation tasks in order to minimize the system cost [91] [92].

- 5) Task replication: Task replication includes deciding both the number of replicas for a task and when and where the task replicas are executed. The completion time of a task is decided based on the first response among the task replicas. Task replication is used to improve both the delay performance and reliability in case of device or network failures [93].

- 6) Service placement/migration: Service placement problem includes mapping the application modelled as a set of services (monolithic, inter-dependent, connected graph) to different devices in the network while considering the different constraints [17]. Service migration, on the other hand, includes deciding where to migrate the application services for better placement in case of user mobility, network/device failures, or to achieve better QoS [15].

- 7) Virtual machine placement/migration: Virtual machine placement/migration includes deciding the deployment and migration of traditional virtual machines, docker container, or other lightweight virtualization technology [16] on edge devices considering the environment constraints. This problem is considered for the cases where the services and application tasks deployed on the edge devices are assigned to virtual machines or containers [94].

- 8) Frequency scaling: CPU frequency of an edge device is another resource that can be adjusted based on the application and system objective. The frequency scaling problem is to decide the CPU frequency of an edge device used to execute the computation task under the system constraints [95] [96].

6.1.2. Data/Storage scheduling

This category includes problems related to the scheduling of data and storage resources at the devices. This category of problems considers different aspects of data, including caching, delivery, replication, and offloading.

1) Data caching: Data caching includes deciding the placement of data at the devices to minimize data access delay and system cost while subject to system constraints including storage capacity [97] [98]. Caching problem considers which data need to be cached at which devices. The caching decision depends on several factors such as the popularity of data, network traffic, etc. Many existing works also consider aspects including request dispatching, data delivery, and cache replacement while modelling the cost and objective for caching problems.

2) Data delivery/dissemination: Data delivery/dissemination problem considers that given the data placement, decide which data or data segments need to be delivered and by which device to achieve maximum performance while subject to system constraints [99] [100] [101]. As data delivery from multiple devices can lead to conflicting transmissions, the problem can also include the time the data is to be delivered to avoid interference.

3) Data replication: Data replication includes deciding the placement of data replicas at multiple devices to reduce the data access delay or optimize other performance metrics subject to system constraints, including storage capacity [102] [85] [103]. Data replication problem can include several decisions, including which data is to be replicated, when to create the replica, how many replicas need to be created, and where to place the data replicas.

4) Data offloading: Similar to computation offloading, as the computation tasks are moved from an IoT device or edge device to another edge device, the corresponding data need to be offloaded too. Data offloading problem includes deciding when and where a particular data is offloaded while considering the system constraints [104] [105] [92]. Besides offloading data corresponding to computation tasks, data can also be offloaded to other devices for other use cases such as aggregation, transfer, etc.

6.1.3. Network scheduling

This category includes problems related to network resources such as bandwidth, wireless channels, etc. These problems are considered depending on the network model, i.e. devices can have shared bandwidth channel, orthogonal channels, or support frequency reuse.

1) Bandwidth allocation: Bandwidth allocation includes deciding the amount of network bandwidth used for transmitting the data or offloading computation tasks. Many works usually consider bandwidth allocation jointly with other problems such as computation offloading, caching, etc. This problem is solved for scenarios where the network bandwidth is shared among multiple users/devices [106] [89].

2) Channel selection/Spectrum allocation: Channel selection/Spectrum allocation includes deciding which chan-

nel is used for transmitting the data. This problem is considered for case where the IoT device (or an edge device) is connected to different edge devices using multiple orthogonal channels [107] [108] [109]. [In recent years, there have been some works that have considered non-orthogonal multi-access \(NOMA\)-enabled multi-access edge computing networks that also involve spectrum allocation \[110\] \[111\] \[112\].](#)

3) Spectrum sharing: Spectrum sharing includes allocating the frequency spectrum among multiple users while considering co-channel interference and other cost constraints. The use of a dedicated spectrum among users can lead to inefficient performance. Therefore, spectrum sharing considers frequency reuse and enables the use of underutilized spectrum to improve spectrum efficiency [106].

4) Routing path selection: Some existing works consider multi-hop offloading, where the devices that are connected to each other using a multi-hop path can use different routing paths to transmit the data. Routing path selection includes deciding the routing path used to transmit the data between devices while considering the network constraints [113] [114].

5) Radio-access technology (RAT) selection: RAT selection includes deciding the radio-access technology, from a set of different radio-access technologies, to connect the end devices to edge devices. [109]. This problem is considered in the case of 5G MEC, where the end devices can be connected to MEC server/edge device using different radio-access technologies.

6) Power control: Power control includes deciding the transmission power used by the device to transmit the data subject to system constraints [95] [115] [116]. The value of transmission power is used for calculating the data rate of transmission and affects the signal-to-interference plus noise (SINR) ratio. [Power control problem is especially considered for NOMA-enabled MEC networks \[117\] \[118\].](#)

6.1.4. Miscellaneous resource scheduling

This category includes problems for scheduling and trading different types of resources, including energy and network slices, that are not covered in the previous three categories.

1) Energy scheduling: Energy scheduling includes deciding the amount of energy generated, consumed, stored, and shared between different devices while considering the system and network constraints. [119] [120]. This problem is often considered for self-powered networks with renewable energy generation capabilities.

2) Network Slicing: Network slicing includes deciding which resources are allocated to each network slice to maximize the overall utility while subject to system constraints. Network slicing problem is considered for 5G MEC network where multiple logical networks, i.e. network slices, can run on a common physical infrastructure [121].

3) Resource trading: Resource trading includes deciding the price/incentive to share different resources such

as computation, storage, data, or other services among different stakeholders (users, devices, service providers, etc.) while considering the system constraints [8] [122]. Since there can be multiple candidate devices (both from provider and receiver end), this problem can also include selecting which devices are involved in the trading/sharing of resources.

6.2. Collaborative scheduling problems

This category includes problems that consider joint scheduling of computation, data/storage, and network resources. These problems include jointly scheduling different independent non-collaborative scheduling problems discussed previously. While collaborative scheduling can also include joint scheduling of energy resources, we have not included it to make the categorization concise as computation, data/storage, and network are often the main resources being scheduled. One of the main challenging issues in proposing a solution for such collaborative scheduling problems is to consider complex dependency among different variables.

1) Joint computation and data/storage: Joint computation and data/storage problem includes making joint scheduling decisions on computation and data/storage resources. Many works have studied the problems of jointly optimizing the computation offloading and caching decision [123] [124] [125] [72]. The work in [123] solved the problem of joint computation offloading and caching to maximize the long-term reward for NOMA-enabled MEC. Another work in [72] also studied the joint computation offloading and caching problem for NOMA-enabled MEC. The problem of dynamic task offloading and caching has also been studied for MEC-enabled cellular networks in [124]. Joint computation and caching problem has also been studied for hybrid mobile/cloud edge computing environment considering the constraints on energy consumption, computation capability, and caching capacity.

2) Joint computation and network: Joint computation and network problem includes making joint scheduling decisions on computation and network resources. Existing works have studied several problems optimizing both the computation offloading and bandwidth allocation or spectrum allocation decision [126] [127] [128]. There are also some existing works that have considered joint task allocation and network flow scheduling problems in [36] [34] [35]. One important direction of works within this sub-category is jointly scheduling computation and network resources for NOMA-assisted MEC. Several works [110] [111] [112] [129] have studied the problem of jointly optimizing computation and network resources for NOMA-assisted multi-access mobile edge computing. Another work in [117] studied joint computation offloading, power control, and spectrum allocation for NOMA-based 5G MEC network that also considers pricing of resources to maximize the utility. The work in [118] studied the power allocation problem for NOMA-enabled MEC for Internet of Vehicles scenario to minimize the delay during offloading. Joint user allocation

and power control problem has also been studied for 5G NOMA-based MEC systems in an existing work [130].

3) Joint data/storage and network: Joint data/storage and network problem includes making joint scheduling decisions on storage and network resources. Existing works such as [131] [132] have studied the joint caching and spectrum allocation problem. The work in [131] studied the joint caching and spectrum allocation problem for MEC-enabled adaptive-bitrate video streaming application. Another work in [132] studied the problem of jointly optimizing the decision for data caching, spectrum allocation, and power control for NOMA-enabled MEC environment.

4) Joint computation, data/storage and network: Joint computation, data/storage, and network problem includes making joint scheduling decisions on computation, storage, and network resources. Several works in literature have studied problems jointly optimizing the variables related to different resources [108] [96] [133] [116]. The work in [108] studied the joint computation offloading, spectrum allocation, and caching problem for MEC and UAV-assisted vehicular networks. The work in [96] solved the joint computation offloading, caching, and spectrum allocation problem considering the total revenue for MEC network. The joint scheduling problem has also been studied for full-duplex enabled small cell network with MEC. Another work in [134] studied the joint computation offloading, caching, and power control problem NOMA-enabled MEC network to achieve minimum computation latency.

7. Solution approaches

Several solutions have been proposed in the literature for the different problems related to DRS in edge computing. This survey broadly classifies the solution approaches in four categories, i.e. machine learning, economic theory, online/approximation, and bio-inspired solution.

7.1. Machine learning

Machine learning approaches, including supervised learning, unsupervised learning, and reinforcement learning, have been applied for several resource scheduling problems as investigated in related surveys [20] [19] [19]. DRS problems have often been solved in the literature using multi-agent reinforcement learning (MARL). MARL involves multiple agents collaboratively interacting and learning using the feedback from the dynamic environment. Each agent uses a policy to choose an action from the action space based on the current environment state and receives the reward. Different algorithms have been proposed in the literature to learn the policy for taking action [135]. There are many challenging issues to be addressed while proposing an algorithm, such as complexity due to large state space and number of agents, stability, nonstationarity, etc.

One of the prevalent problems addressed using MARL is computation offloading. Several variants of the solution have been proposed considering different system

models, constraints, and objectives. The work in [119] proposed an advantage actor-critic algorithm (A2C) to make offloading and energy allocation decisions for multiple non-cooperative mobile devices, where each mobile device only has a partial observation. The work in [126] proposed MARL solution for joint computation offloading and bandwidth allocation problem. [The work in \[72\] proposed bayesian learning automata based multi-agent Q-learning algorithm to solve the joint computation offloading and caching problem in NOMA-enabled MEC.](#) A multi-agent deep reinforcement learning (MADRL) algorithm has also been proposed in [136] for computation offloading problem, where each server is equipped with Deep Q-Network (DQN) agent to select associated users that can offload tasks. Another work in [137] has also proposed a MARL solution using DQN agents for task scheduling. The work in [138] proposed a deep deterministic policy gradient (DDPG) algorithm to learn dynamic computation offloading policy at each user independently without prior knowledge of the system. MARL has also been used in combination with other machine learning algorithms, i.e. short-term memory (LSTM) and bidirectional recurrent neural network (BRNN), to provide better computation offloading solution in [139]. Several works [140], [141], [142] have proposed MADRL solution, using the popular algorithm Multi-Agent Deep Deterministic Policy Gradient (MADDPG), for computation offloading problem. Joint computation offloading and resource allocation problems have also been studied. MADRL solution is proposed in [141] to solve the joint channel selection and computation offloading problem. The joint problem of path planning and computation offloading has been solved using the MADRL solution in [142] for a multi-UAV environment.

Caching is another critical problem that has been studied in several existing works. MARL has been used in [98] to solve cooperative content caching problem, assuming no prior knowledge of content popularity and users' preference. Another work in [143] has proposed the use of Federated deep reinforcement learning to enable decentralized cooperative edge caching. A joint computation offloading and caching problem has also been studied in [123] where an exhaustive search is used to determine the caching decision and multi-agent Q-learning is used to solve the computation offloading problem. The work in [144] proposed the use of a cognitive agent where multiple agents collaborate and use LSTM to predict user's mobility and service type to improve the cache hit ratio.

Other resource allocation problems have also been solved using MARL. The work in [109] uses a multi-agent Q-learning algorithm to solve the computation offloading problem where each user also independently solves the resource allocation problem of determining the transmit power level, the sub-channel and the radio access technology. Another work [121] has studied the network slicing problem and proposed a decentralized deep reinforcement learning method to orchestrate end-to-end resources. The energy dispatching problem has also been studied

and solved using multi-agent meta-reinforcement learning framework [120].

Various solutions have also been proposed in literature considering different application domains. Several works [108], [145], [146] have solved the problem for vehicular networks. MADRL solution has been proposed for computation offloading in vehicular networks in [146]. The work in [108] used the MADDPG algorithm to solve the problem of the spectrum, computation, and caching resource allocation in UAV-assisted vehicular networks. MARL solution has been proposed for distributed real-time service scheduling in the vehicular network while considering requirements including real-time processing, heterogeneity, and vehicle mobility. Another work in [147] proposed MADRL-based asynchronous advantage actor-critic algorithm for energy scheduling in edge computing-enabled microgrid. Distributed deep reinforcement learning solution has also been proposed to decide the offloading of functions in an edge-enabled small-cell 5G network. MARL has also been used in UAV networks for jointly optimizing user, power level and subchannel selection.

Drawbacks: Machine learning methods involve heavy training overhead that leads to significant computation, memory, storage, and time cost. There are also large number of hyper-parameters that need to be considered while designing the solution. In particular, there are also some specific open challenges with regards to MARL solutions proposed for DRS. First is nonstationarity in environment, which implies that reward received by an agent for a certain action depends on the actions taken by other agents. Second is scalability, which implies that the increase in the number of agents exponentially increases the action space, leading to consequent increase in the compute resource and memory requirement for training. Third is partial observable environment, which implies that each agent does not have global knowledge and observes different instances of state. Additionally, reward function used for each agent should be designed appropriately to achieve the optimal solution for scheduling.

7.2. Economic theory

Economic theory approach includes use of game theory and auction theory to model the interaction among entities (devices, users, and service consumers and providers) using market economics [8] [29]. Existing works in literature have been modeled using different games including minority game, potential game, submodular game, etc.

Minority game has been used for solving the computation offloading problem in several existing works [91], [148], [149], [150], [151]. In a minority game, the odd number of players decide between two actions and only the players on the minority side win. Minority game has been used in [91] to solve the server activation problem during computation offloading. Another work [148] used both minority game and reinforcement learning to solve the server activation problem. The work in [149] solved the computation of-

floading problem for heterogeneous tasks with incomplete information using minority game.

Potential game is another popular approach for modelling the DRS problems. The work in [152] formulated the multi-user computation offloading problem as an exact potential game with pure-strategy Nash Equilibrium Point. This work further used the machine learning approach and proposed a fully distributed computation offloading algorithm [152] which can converge without information exchange between devices. Another work in [153] has also used a potential game to model the computation offloading problem in a small-cell 5G MEC network. The potential game has also been used to solve user allocation problem in [89]. User allocation problem has been solved using the game-theoretical approach in many other works [89], [90].

Another game-theoretic approach is using prospect theory to solve the offloading problem. The work in [104] solved the data offloading problem using prospect theory while considering the human awareness, inherent cognitive biases and behavioral characteristics of devices. Another work [105] also used prospect theory to model the data offloading problem. The dynamic sequential game also has been used to solve multi-user computation offloading problem [154]. Another work modelled the computation offloading problem as a submodular game and proved the existence of at least one Pure Nash Equilibrium point. Game theory has also been used to solve the partial computation offloading problem [155]. Congestion game has also been used in [156] to give a distributed solution for service placement in mobile edge computing. Another approach used the Stackelberg game to model the interaction between edge cloud and users where edge cloud sets the price and each user locally makes offloading decision [157]. A strategic game has also been used in [158] to give a distributed computation offloading solutions for multiple mobile users.

Game theory has also been used to solve the caching problem. The work in [131] used the Stackelberg game to solve the joint caching and adaptive bitrate streaming problem, i.e., both storage allocation and adjusting the transmission bitrate based on the network condition. Another work in [159] also used the Stackelberg game for solving the caching problem in heterogeneous networks for mobile users with multihoming access. Caching problem has also been solved using mean-field game theory in [160].

The joint resource allocation problem has also been solved using game-theoretic approaches. The work in [95] used game theory to solve the joint computation offloading and resource allocation problem, including offloading decision, CPU capacity adjustment, transmission power control, and network interference management. The work in [115] uses game theory to model the problem of transmission power control in an interference-aware multi-user scenario. A two-tier matching game has also been used in [122] to solve the computing resource allocation and service pricing problem in small-cell networks.

Several works have also combined game theory with machine learning algorithms. The work in [161] proposed deep reinforcement learning-based game for solving computation offloading problem. Another work [162] combined mean-field game theory with deep reinforcement learning for solving the task allocation problem. Game theory has also been integrated with Generalized Adversarial Imitation Learning for solving computation offloading problem with partial observations. Another work in [163] proposed a hierarchical game-theoretic and reinforcement learning framework that includes a coalition formation game played across the network and a non-cooperative subgame played inside the coalition. Game theory has been used for solving DRS problems in different application domains including UAV network [164], vehicular network [165], [166], and blockchain-empowered edge computing for Industry IoT [167].

Few works have also used auction theory to provide a semi-distributed solution. The work in [168] uses Vickrey-Clarke-Groves auction to propose a semi-distributed solution for the computation offloading problem considering the dynamic energy harvesting process and task arrivals. Another work [169] also uses auction-based implementation to provide a semi-distributed solution for the joint computation offloading and multiuser scheduling (channel selection) problem. The work in [170] used the iterative combinatorial auction to provide distributed solution service placement problem.

Drawbacks: One main issue with using game theoretic approaches in economic theory for distributed scheduling is that there is an additional traffic overhead resulting from data exchange between different players (edge devices) in a game [29]. Furthermore, it is often assumed in game theoretic approaches that different players are aware of the cost/utility functions of other players. There are also other limitations with game theory such as mutual satisfactory solution with Nash equilibrium among different players is not necessary a global optimum solution, and continuous iterations are required to obtain the Nash equilibrium [25] leading to increase in computation complexity.

7.3. Online/approximation Algorithms

Online/approximation algorithms include the use of various distributed convex optimization techniques proposed in the literature. A characteristic feature of these solutions is formulating the DRS problems into a specific model to use the distributed convex optimization technique. Therefore, these solutions often involve complex mathematical modelling of the problems. Some of the popular techniques used to solve DRS problems in edge computing include alternating direction method of multipliers (ADMM), Lyapunov optimization, Markov approximation, etc.

ADMM has been used in several existing works to solve the various DRS problems, including computation offloading, caching, and resource allocation. For example, the work in [125] used ADMM to propose a dis-

tributed algorithm for solving the joint computation offloading and caching problem. ADMM has also been used in [96] for solving joint computation offloading, caching, and resource allocation problem. Another work [171] proposed two different distributed algorithms, i.e. subgradient method with dual decomposition and distributed alternating direction method of multipliers via variable splitting (ADMM-VS) for solving computation offloading problem among neighbouring devices. ADMM has been used in [116] for joint computation offloading, caching, spectrum allocation and power control. ADMM has also been used for blockchain-based video streaming systems in [107] for joint optimization of computation offloading, block size, and spectrum allocation.

Besides ADMM, there are other techniques too for solving DRS problems in edge computing. The work in [172] used the Markov approximation approach to enable fully distributed placement of virtual machines on edge devices. Another approach for a distributed solution is Block Successive Upper-bound Minimization (BSUM) method proposed for solving joint computation offloading and caching problem [133]. A decentralized solution has been proposed for joint computation offloading and caching using an approach based on Gibbs sampling [124]. [Another work in \[134\] used BSUM to solve the joint computation offloading, caching and power control problem for NOMA-enabled MEC networks.](#)

Several works have also proposed distributed online optimization solutions for DRS in edge computing. The work in [51] proposed distributed online learning solution for computation offloading problem that uses distributed approximate graph matching method. Distributed online learning solution has also been proposed for content placement and delivery in [97] by using stochastic gradient descent (SGD). An asynchronous distributed online optimization for computation offloading and task migration decision has been done in [173] using nonlinear problem transformation. Another distributed online optimization solution has been provided for computation offloading in [174], where there is no a priori knowledge of future task arrivals and network conditions beyond the immediate neighbours. A distributed online optimization solution has been proposed for computation offloading in [175], where the data can be partitioned, processed, and dispatched among multiple servers. A distributed online optimization solution has also been proposed for caching problem in [176]. Few works have also considered multi-hop computation offloading. The work in [113] proposed a distributed algorithm for multi-hop computation offloading, which includes jointly making the decision on server selection and traffic routing. Another work in [114] also proposed distributed online optimization solution for multi-hop computation offloading.

Some works have also proposed distributed and parallel implementation of a solution by decomposing the problem into subproblems. One example is work in [177] where the computation offloading problem in multicellular MEC is

decomposed among subproblems across different small cell base stations with limited signalling between small cells and cloud. Another work in [178] decomposed the problem of joint optimization of computation offloading, UAV trajectory, and user transmit power. The work in [87] has decomposed the long-term joint computation offloading and energy management problem to solve the problem in a distributed manner without requiring future system statistical information. [The work in \[110\] leveraged the layered structure to decompose the joint computation offloading and time allocation problem for NOMA-assisted MEC into sub-problems that are solved using convex optimization technique.](#)

Drawbacks: Online/approximation algorithms require complex mathematical modelling of the problem that is not feasible in many practical scenarios. Many online solutions adopt heuristic approaches to solve the decomposed problem that often achieve locally optimal solution. Furthermore, these solutions do not have a performance guarantee.

7.3.1. Bio-inspired Algorithms

Bio-inspired algorithms include solutions based on the biological phenomena of plants, animals, and group behavior of animals such as birds, ants, etc. Bio-inspired solutions have been extensively used for multi-agent systems [3] [179]. There are few works on using bio-inspired solutions for DRS in edge computing. The work in [180] proposed a service placement solution for the decentralized cloud with dynamic network topology using a bio-inspired algorithm based on growth dynamics of hyphae (the root-like structures of some species of fungi). The system model used in this paper can be applied for peer-to-peer networks, edge-cloud systems. Another work in [181] proposed a multicellular embryonic fog platform for service delivery based on embryonic cellular development.

There are existing works using bio-inspired algorithms in related fields such as cognitive IoT, wireless sensor networks, etc. The work in [182] proposed a solution for cluster formation for cognitive IoT using a reaction-diffusion mechanism. Another work in [183] proposed a solution for decentralized radio access solution in cognitive radio based on social foraging swarming models in birds. A distributed solution has been proposed for clustering in wireless sensor networks using firefly-inspired algorithms [184]. Another related work is for distributed Transmission Power Control in a wireless body area network using the Cucker-Smale model [185]. Since the system model in these works is quite similar to edge computing, the bio-inspired solutions proposed in these works can also be extended for DRS in edge computing.

Drawbacks: Bio-inspired solutions often converge to a local optimum solution and the performance of the solution cannot be guaranteed. There are also very few works that have explored bio-inspired algorithms for solving DRS problem in edge computing. Most existing works have

leveraged bio-inspired algorithms for DRS in different contexts with homogeneous devices.

7.4. Discussion

Table 3 shows a summary of the investigation of existing works based on the type of solution approach. We have investigated different existing works for each solution approach and identified the resources being scheduled, type of problem and performance metric considered in work. For the solution approach column, we have included the specific approach such as MARL, MADRL, or DRL within the Machine Learning approach and Game Theory, Game Theory and ML, or Auction Theory within the Economic Theory. For the resources being scheduled column, we have categorized in terms of four main resources: computation, data/storage, network, and energy. The specific problem being studied corresponding to the resource is mentioned in the problem column.

We have made few observations based on the investigation of existing works in Table 3:

- Existing works have studied different types of problems including computation offloading, caching, energy scheduling, bandwidth allocation, power control, etc., while considering different system model and constraints. Some of the problems identified in this survey such as data dissemination, data replication, and spectrum sharing have not yet been fully explored. There are also few recent works considering resource sharing and trading among different devices considering privacy requirements.
- One of the main problem being studied for DRS in edge computing is scheduling computing resources, with computation offloading being the most popular problem. Among the different unique issues of DRS highlighted in Section 2.2, heterogeneity and network dynamics and wireless interference have received the most attention. Since the edge computing environments are especially characterized by limited and unstable network bandwidth, joint computation and network resource is one of the unique problems of DRS in edge computing. NOMA-enabled MEC environment, in particular, involves studying joint computation and network scheduling problems [110] [111] [112] [129]. Some works have also combined caching problem together to jointly schedule computation, data/storage, and network resources.
- Several variants of the solution approaches have been proposed in existing works. In addition, many works have also integrated different approaches such as DRL and game theory or auction theory. However, very few works are using the bio-inspired solution for solving DRS problems in edge computing. The main reason being that most bio-inspired solutions are usually based on the assumption of homogeneous

devices. In contrast, edge computing is characterized by heterogeneous devices with dynamic networks and workloads.

- Several performance metrics have been considered in existing works, including delay, energy, system cost, bandwidth consumption, and utility. As pointed out earlier in Table 2, the target of resource scheduling problems in edge computing is to minimize latency and energy consumption. It is reflected in existing works as most works consider either delay or energy or both as part of the objective function. Although some existing works have considered optimizing the system utility, most works do not focus on traditional performance metrics of load balancing and network utilization.

8. Future Directions

The investigation done in this survey shows that many issues can be further explored. This section describes some of the future directions for DRS in edge computing.

8.1. Integrated Platform

Many different platform solutions have been proposed for DRS in the literature. However, as investigated, most of these platform solutions are either application-specific or problem-specific. One main opportunity for future work is to propose integrated platforms that consider distributed scheduling of different types of resources in edge computing. One issue to consider is designing the architecture for the integrated platform. Most recent works on edge computing use microservice architecture for the platform, where the different resource scheduling solutions are deployed as microservices on the devices. Another issue to consider is to design the interfaces to enable interactions among different components in the architecture and with the application users. The integrated platform should also include middleware to provide an abstraction for the application users. Many middleware solutions such as Emma [194], EdgeDrone [195], etc. have been proposed for edge computing that can serve as starting point.

8.2. Realistic Testbeds and Simulation Toolkits for Evaluation

Some recent works have created testbeds to test the efficacy of proposed solutions for edge computing [196] [197] [198]. However, these testbeds have not been designed considering the DRS solutions in edge computing. There is an opportunity to design testbeds with public access that can enable the evaluation of DRS solutions. Another approach for evaluating the proposed solutions is using simulation toolkits since it can be expensive and difficult to develop testbeds for all scenarios. There are few works in literature that have proposed simulation toolkits such as iFogSim [199], EdgeCloudSim [200], FogNetSim++ [201], etc. to

Solution Approach	References	Resources being scheduled	Problem	Objective
MARL	[139] [186] [187]	Computation	Computation Offloading	Delay, Energy
	[137] [145] [98]	Computation Computation Data/storage	Task scheduling Task allocation/scheduling Caching	Delay Delay Hit rate, Delay
	[120] [109]	Energy Joint computation and network	Energy scheduling Computation Offloading, Power Control, RAT selection, and Channel Selection	Energy, Accuracy Delay, Energy
	[188]	Joint computation and network	User allocation, Power Control, and Channel Selection	Throughput, Power
	[126]	Joint computation and network	Computation offloading and Bandwidth allocation	Delay
	[127]	Joint computation and network	Computation offloading and Channel selection	Delay, Energy
	[123]	Joint computation and data/storage	Computation offloading, User allocation, and Caching	Energy
	[72]	Joint computation and data/storage	Computation offloading and Caching	Delay, Energy
	[119]	Joint computation and energy	Computation offloading and Energy scheduling	Delay, Energy
	[108]	Joint computation, data/storage, and network	Computation offloading, Caching, and Spectrum allocation	Delay, QoS
MADRL	[146] [142] [136] [140]	Computation Computation Computation	Computation offloading Computation offloading Computation offloading	Delay Delay, Energy Delay, Energy, Task failure rate
	[147] [138] [189]	Energy Computation Computation	Energy scheduling Computation offloading Computation offloading	Energy Delay, Power Energy, Traffic drop rate
	[143]	Data/storage	Caching	Delay, Hit rate, Traffic load
	[144] [121] [141]	Data/storage Network Joint computation and network	Caching Network slicing Computation offloading and Channel selection	Delay, Hit rate QoS Delay, Channel access success rate
Game Theory	[148] [150] [115]	Computation Computation Computation	Computation offloading Computation offloading Computation offloading and Power control	Energy Delay, QoE Delay, Energy
	[158] [152] [165] [166]	Computation Computation	Computation offloading Computation offloading	Delay Delay, Energy
	[154] [155] [164] [163] [153] [167] [89]	Computation Computation Computation	Computation offloading User allocation	Delay, Energy Number of users, System cost
	[156] [151]	Computation Computation	Service placement Computation offloading	Delay Delay, Energy, Transmission power
	[157] [90]	Computation Computation	Computation offloading User allocation	Delay, Price Number of users, Systems cost
	[91]	Computation	Computation offloading and Server activation	Delay, Energy
	[122]	Computation	Computation offloading and Resource trading	Delay, Energy
	[190]	Computation	Computation offloading and Resource trading	Delay, Energy, Price
	[191]	Computation	Task allocation/scheduling and Resource trading	Utility

Table 3: Summary of existing works

Solution Approach	References	Resources being scheduled	Problem	Objective
Game Theory	[104] [92] [105] [159] [117]	Data/storage Data/storage Data/storage Joint computation and network	Data offloading Data offloading Caching Computation offloading, Power control, allocation, spectrum	Delay, Energy User utility Utility Delay, Energy, Utility
	[129]	Joint computation and network	Computation offloading, Power control, and spectrum allocation	Delay, Energy
	[95]	Joint computation and network	Computation offloading, Power Control, Channel Selection, and Frequency scaling	Delay, Energy
	[192]	Joint computation and energy	Computation offloading and Energy scheduling	Delay, Energy
	[131]	Joint data/storage and network	Caching, Spectrum allocation, and Bitrate adaption	Cache hit ratio, Throughput
	[132]	Joint data/storage and network	Caching, Spectrum allocation, and Power control	System cost
Game theory and ML	[161] [162] [142]	Computation Computation Computation	Computation offloading Task allocation Computation offloading	Delay Delay Delay
	[168] [170] [169]	Computation Computation Joint computation and network	Computation offloading Service placement Computation offloading and Channel selection	Delay, Energy System cost Delay, Energy
Online/approximation	[173] [177] [113] [114] [178]	Computation Computation Computation	Computation offloading Computation offloading Computation offloading and Trajectory optimization	Delay, Energy Energy Delay, Energy
	[171]	Computation	Computation offloading	Delay, Power efficiency
	[174] [51] [172] [176] [97]	Computation Computation Data/storage Data/storage	Computation offloading Service placement Caching Caching	Delay System cost Cost Cache hit ratio, Cost
	[175] [106]	Data/storage Network	Data offloading Bandwidth allocation and Spectrum sharing	Delay, Energy Energy
	[118]	Joint computation and network	Power control and Spectrum allocation	Delay
	[107]	Joint computation and network	Computation offloading, Spectrum allocation, Frequency scaling, and Adaptive block size	Delay, Energy, and Bandwidth consumption
	[110] [111] [112] [193]	Joint computation and network Joint computation and network	Computation offloading and Power control Task allocation/scheduling and Bandwidth allocation	Delay, Energy Delay, Revenue
	[124] [125]	Joint computation and data/storage	Computation offloading and Caching	Delay, Energy
	[87]	Joint computation and energy	Task scheduling and Energy scheduling	Throughput, Fairness,
	[133]	Joint computation, data/storage, and network	Computation offloading, Caching, and Bandwidth allocation	Delay, Throughput, Cache hit ratio, and Bandwidth saving
	[134]	Joint computation, data/storage, and network	Computation offloading, Caching, and Power control	Delay, Energy
	[116]	Joint computation, data/storage, and network	Computation Offloading, Caching, Power control, and Bandwidth allocation	Delay, Revenue, Cache hit ratio, and Power
	[96]	Joint computation, data/storage, and network	Computation offloading, Caching, Spectrum allocation and Frequency scaling	Delay, Utility

Solution Approach	References	Resources scheduled	being	Problem	Objective
Bio-inspired	[180]	Computation		Service placement	Delay, System cost
	[182]	Network		Spectrum allocation	Delay, Throughput
	[181]	Data/storage		Data delivery	Resilience

Table 3 Continued: Summary of existing works

evaluate the proposed solutions for resource scheduling in edge/fog computing environments. This leads to another opportunity for future works to build simulation toolkits that can provide benchmark models and solutions specifically for DRS solutions. This would help researchers to have an easier and fairer comparison of the proposed solutions. Future works can also consider providing hybrid testbeds, including both real devices and emulation platforms, to evaluate the scalability of proposed solutions in real environments.

8.3. Collaborative Scheduling

Many existing works on DRS provide a solution for managing individual resources. Few works have explored joint optimization of computation offloading, caching, or resource allocation problems [95] [123]. One approach usually followed in many existing works is to solve the joint problem by separating the joint problems and using uni-directional invocation among schedulers of different resources. For example, the computation scheduler invokes the network scheduler to decide the bandwidth allocation. Such a solution would lead to inefficient performance as computation scheduler decision is not affected by network flows. There is an opportunity to propose collaborative scheduling solutions where the schedulers for different resources such as computation, network, and storage cooperate with bi-directional invocations. Collaborative scheduling solutions have been shown to provide performance benefits in terms of reduced completion time and network traffic for centralized resource scheduling [34] [35][88]. There are additional issues to propose DRS solution, as highlighted in Section 2, that have not been considered for collaborative scheduling.

8.4. Complex System Model

Existing works have considered some of the unique issues pointed out in Section 2 such as heterogeneity, network dynamics, incentive mechanisms, etc. However, there are many issues emerging due to complexity in system model including heterogeneous resource access and availability, application adaptability, and other aspects related to data and network in edge computing have not been well-explored in the literature. Few solutions have considered data characteristics including multiple sources of data, age of information, and value of information. Network characteristics including hierarchical network, mesh-network architecture, and heterogeneous

resource access are also not considered in many existing works related to DRS. There are also future opportunities to propose distributed solutions considering different system models and constraints including multi-hop network topology, streaming applications, and stateful applications.

8.5. Integration with Emerging Technologies

Another opportunity for future works is to model and solve the DRS problems considering integrating edge computing with emerging technologies including Blockchain, AI, and future 5G/6G networking technologies. DRS solutions for edge computing-enabled blockchain and AI require considering additional parameters. For example, there are recent works on integrated edge computing and blockchain that consider different resource scheduling problems, including mining offloading, transaction size, transaction price, etc. [69] [202]. However, many of these works assume centralized controller and issues specific to DRS in edge computing have not been well-explored. Similarly, for Edge AI resources need to be managed while jointly considering the training and inference of AI models. For example, during federated learning, parameters such as aggregation frequency, incentive mechanisms, selecting local worker nodes, etc., need to be optimized [203] [204]. There are also opportunities to propose novel DRS solutions while integrating edge computing with 5G, 6G, or other future network technologies.

9. Conclusion

This survey investigates the existing works to provide a holistic view of the different aspects of DRS in edge computing, including the challenging issues, application scenarios, motivations, taxonomy, and future directions. Many unique challenging issues have been identified to enable DRS in edge computing, including heterogeneity, network dynamics and wireless interference, device access and resource availability, multiple stakeholders, etc. We describe several emerging application scenarios, including autonomous vehicles, real-time video analytics, Industry IoT, etc., that need DRS in edge computing. The different motivations for using DRS in edge computing are described, including low-latency, high scalability, fault tolerance, collaboration among different devices, etc. to enable emerging application scenarios in edge computing. A taxonomy is provided to classify the existing works for

DRS in edge computing from three perspectives, i.e. system, problems, and solutions approach. We also summarize and classify the different works based on four solution approaches investigated in this survey: machine learning, economic theory, online/approximation, and bio-inspired algorithm. Finally, several future directions have been identified such as integrated platform, realistic testbed and simulation toolkit for evaluation, collaborative scheduling, integration with emerging technologies, etc., to help advance state-of-the-art for DRS in edge computing.

Acknowledgement

The work described in this paper was supported in part by the Hong Kong (HK) Research Grant Council (RGC) General Research Fund with project code PolyU 15217919, in part by the HK RGC Research Impact Fund with project code R5060-19, and in part by the HK RGC Collaborative Research Fund with project code C5018-20GF.

References

- [1] W. Shi, J. Cao, Q. Zhang, Y. Li, L. Xu, Edge computing: Vision and challenges, *IEEE internet of things journal* 3 (5) (2016) 637–646.
- [2] Y. Sahni, J. Cao, S. Zhang, L. Yang, Edge mesh: A new paradigm to enable distributed intelligence in internet of things, *IEEE access* 5 (2017) 16441–16458.
- [3] L. E. Parker, Distributed intelligence: Overview of the field and its application in multi-robot systems., in: *AAAI Fall Symposium: Regarding the Intelligence in Distributed Intelligent Systems*, 2007, pp. 1–6.
- [4] Y. Mao, C. You, J. Zhang, K. Huang, K. B. Letaief, A survey on mobile edge computing: The communication perspective, *IEEE communications surveys & tutorials* 19 (4) (2017) 2322–2358.
- [5] N. Abbas, Y. Zhang, A. Taherkordi, T. Skeie, Mobile edge computing: A survey, *IEEE Internet of Things Journal* 5 (1) (2017) 450–465.
- [6] W. Yu, F. Liang, X. He, W. G. Hatcher, C. Lu, J. Lin, X. Yang, A survey on the edge computing for the internet of things, *IEEE access* 6 (2017) 6900–6919.
- [7] P. Mach, Z. Becvar, Mobile edge computing: A survey on architecture and computation offloading, *IEEE Communications Surveys & Tutorials* 19 (3) (2017) 1628–1656.
- [8] N. C. Luong, P. Wang, D. Niyato, Y. Wen, Z. Han, Resource management in cloud networking using economic analysis and pricing models: A survey, *IEEE Communications Surveys & Tutorials* 19 (2) (2017) 954–1001.
- [9] S. Wang, X. Zhang, Y. Zhang, L. Wang, J. Yang, W. Wang, A survey on mobile edge networks: Convergence of computing, caching and communications, *Ieee Access* 5 (2017) 6757–6779.
- [10] K. Toczé, S. Nadjm-Tehrani, A taxonomy for management and optimization of multiple resources in edge computing, *Wireless Communications and Mobile Computing* 2018.
- [11] V. Moysiadis, P. Sarigiannidis, I. Moscholios, Towards distributed data management in fog computing, *Wireless Communications and Mobile Computing* 2018.
- [12] C.-H. Hong, B. Varghese, Resource management in fog/edge computing: a survey on architectures, infrastructure, and algorithms, *ACM Computing Surveys (CSUR)* 52 (5) (2019) 1–37.
- [13] J. Ren, D. Zhang, S. He, Y. Zhang, T. Li, A survey on end-edge-cloud orchestrated network computing paradigms: Transparent computing, mobile edge computing, fog computing, and cloudlet, *ACM Computing Surveys (CSUR)* 52 (6) (2019) 1–36.
- [14] L. Lin, X. Liao, H. Jin, P. Li, Computation offloading toward edge computing, *Proceedings of the IEEE* 107 (8) (2019) 1584–1607.
- [15] Z. Rejiba, X. Masip-Bruin, E. Marín-Tordera, A survey on mobility-induced service migration in the fog, edge, and related computing paradigms, *ACM Computing Surveys (CSUR)* 52 (5) (2019) 1–33.
- [16] Z. Tao, Q. Xia, Z. Hao, C. Li, L. Ma, S. Yi, Q. Li, A survey of virtual machine management in edge computing, *Proceedings of the IEEE* 107 (8) (2019) 1482–1499.
- [17] F. A. Salaht, F. Desprez, A. Lebre, An overview of service placement problem in fog and edge computing, *ACM Computing Surveys (CSUR)* 53 (3) (2020) 1–35.
- [18] M. Ghobaei-Arani, A. Souiri, A. A. Rahmanian, Resource management approaches in fog computing: a comprehensive review, *Journal of Grid Computing* 18 (1) (2020) 1–42.
- [19] A. Shakarami, M. Ghobaei-Arani, A. Shahidinejad, A survey on the computation offloading approaches in mobile edge computing: a machine learning-based perspective, *Computer Networks* (2020) 107496.
- [20] X. Wang, Y. Han, V. C. Leung, D. Niyato, X. Yan, X. Chen, Convergence of edge computing and deep learning: A comprehensive survey, *IEEE Communications Surveys & Tutorials* 22 (2) (2020) 869–904.
- [21] Q.-V. Pham, F. Fang, V. N. Ha, M. J. Piran, M. Le, L. B. Le, W.-J. Hwang, Z. Ding, A survey of multi-access edge computing in 5g and beyond: Fundamentals, technology integration, and state-of-the-art, *IEEE Access* 8 (2020) 116974–117017.
- [22] F. Saeik, M. Avgeris, D. Spatharakis, N. Santi, D. Dechouniotis, J. Violos, A. Leivadidas, N. Athanasopoulos, N. Mitton, S. Papavassiliou, Task offloading in edge and cloud computing: A survey on mathematical, artificial intelligence and control theory solutions, *Computer Networks* 195 (2021) 108177.
- [23] R. A. Dziyauddin, D. Niyato, N. C. Luong, A. A. A. M. Atan, M. A. M. Izhar, M. H. Azmi, S. M. Daud, Computation offloading and content caching and delivery in vehicular edge network: A survey, *Computer Networks* 197 (2021) 108228.
- [24] L. Bréhon-Grataloup, R. Kacimi, A.-L. Beylot, Mobile edge computing for v2x architectures and applications: A survey, *Computer Networks* 206 (2022) 108797.
- [25] Q. Luo, S. Hu, C. Li, G. Li, W. Shi, Resource scheduling in edge computing: A survey, *IEEE Communications Surveys & Tutorials*.
- [26] A. Alwarafy, K. A. Al-Thelaya, M. Abdallah, J. Schneider, M. Hamdi, A survey on security and privacy issues in edge computing-assisted internet of things, *IEEE Internet of Things Journal*.
- [27] Y. Xiao, Y. Jia, C. Liu, X. Cheng, J. Yu, W. Lv, Edge computing security: State of the art and challenges, *Proceedings of the IEEE* 107 (8) (2019) 1608–1631.
- [28] T. L. Duc, R. G. Leiva, P. Casari, P.-O. Östberg, Machine learning methods for reliable resource provisioning in edge-cloud computing: A survey, *ACM Computing Surveys (CSUR)* 52 (5) (2019) 1–39.
- [29] J. Moura, D. Hutchison, Game theory for multi-access edge computing: Survey, use cases, and future trends, *IEEE Communications Surveys & Tutorials* 21 (1) (2018) 260–288.
- [30] K. Krauter, R. Buyya, M. Maheswaran, A taxonomy and survey of grid resource management systems for distributed computing, *Software: Practice and Experience* 32 (2) (2002) 135–164.
- [31] A.-C. Orgerie, M. D. d. Assuncao, L. Lefevre, A survey on techniques for improving the energy efficiency of large-scale distributed systems, *ACM Computing Surveys (CSUR)* 46 (4) (2014) 1–31.
- [32] N. Slamnik-Kriještorac, H. Kremo, M. Ruffini, J. M. Marquez-Barja, Sharing distributed and heterogeneous resources toward end-to-end 5g networks: A comprehensive survey and a taxonomy, *IEEE Communications Surveys & Tutorials* 22 (3) (2020)

- 1592–1628.
- [33] M. D. de Assuncao, A. da Silva Veith, R. Buyya, Distributed data stream processing and edge computing: A survey on resource elasticity and future directions, *Journal of Network and Computer Applications* 103 (2018) 1–17.
 - [34] Y. Sahni, J. Cao, L. Yang, Y. Ji, Multi-hop offloading of multiple dag tasks in collaborative edge computing, *IEEE Internet of Things Journal*.
 - [35] Y. Sahni, J. Cao, L. Yang, Y. Ji, Multi-hop multi-task partial computation offloading in collaborative edge computing, *IEEE Transactions on Parallel and Distributed Systems* 32 (5) (2020) 1133–1145.
 - [36] Y. Sahni, J. Cao, L. Yang, Data-aware task allocation for achieving low latency in collaborative edge computing, *IEEE Internet of Things Journal* 6 (2) (2018) 3512–3524.
 - [37] I. Psaras, Decentralised edge-computing and iot through distributed trust, in: *Proceedings of the 16th Annual International Conference on Mobile Systems, Applications, and Services*, 2018, pp. 505–507.
 - [38] R. Roman, J. Zhou, J. Lopez, On the features and challenges of security and privacy in distributed internet of things, *Computer Networks* 57 (10) (2013) 2266–2279.
 - [39] S. Liu, L. Liu, J. Tang, B. Yu, Y. Wang, W. Shi, Edge computing for autonomous driving: Opportunities and challenges, *Proceedings of the IEEE* 107 (8) (2019) 1697–1716.
 - [40] L. Hobert, A. Festag, I. Llatser, L. Altomare, F. Visintainer, A. Kovacs, Enhancements of v2x communication in support of cooperative autonomous driving, *IEEE communications magazine* 53 (12) (2015) 64–70.
 - [41] A. Marjovi, M. Vasic, J. Lemaitre, A. Martinoli, Distributed graph-based convoy control for networked intelligent vehicles, in: *2015 IEEE Intelligent Vehicles Symposium (IV)*, 2015, pp. 138–143.
 - [42] B. V. Philip, T. Alpcan, J. Jin, M. Palaniswami, Distributed real-time iot for autonomous vehicles, *IEEE Transactions on Industrial Informatics* 15 (2) (2018) 1131–1140.
 - [43] A. M. de Souza, L. A. Villas, A fully-distributed traffic management system to improve the overall traffic efficiency, in: *Proceedings of the 19th ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems*, 2016, pp. 19–26.
 - [44] J. Pan, I. S. Popa, C. Borcea, Divert: A distributed vehicular traffic re-routing system for congestion avoidance, *IEEE Transactions on Mobile Computing* 16 (1) (2016) 58–72.
 - [45] K. Wang, H. Yin, W. Quan, G. Min, Enabling collaborative edge computing for software defined vehicular networks, *IEEE Network* 32 (5) (2018) 112–117.
 - [46] Q. Zhang, Y. Wang, X. Zhang, L. Liu, X. Wu, W. Shi, H. Zhong, Openvdap: An open vehicular data analytics platform for cavs, in: *2018 IEEE 38th International Conference on Distributed Computing Systems (ICDCS)*, IEEE, 2018, pp. 1310–1320.
 - [47] J. Tang, R. Yu, S. Liu, J.-L. Gaudiot, A container based edge offloading framework for autonomous driving, *IEEE Access* 8 (2020) 33713–33726.
 - [48] G. Ananthanarayanan, P. Bahl, P. Bodík, K. Chintalapudi, M. Philipose, L. Ravindranath, S. Sinha, Real-time video analytics: The killer app for edge computing, *computer* 50 (10) (2017) 58–67.
 - [49] Q. Zhang, H. Sun, X. Wu, H. Zhong, Edge video analytics for public safety: A review, *Proceedings of the IEEE* 107 (8) (2019) 1675–1696.
 - [50] F. Loewenherz, V. Bahl, Y. Wang, Video analytics towards vision zero, *Institute of Transportation Engineers. ITE Journal* 87 (3) (2017) 25.
 - [51] C. Ren, X. Lyu, W. Ni, H. Tian, R. P. Liu, Distributed online learning of fog computing under nonuniform device cardinality, *IEEE Internet of Things Journal* 6 (1) (2018) 1147–1159.
 - [52] M. A. Uddin, A. Alam, N. A. Tu, M. S. Islam, Y.-K. Lee, Siat: A distributed video analytics framework for intelligent video surveillance, *Symmetry* 11 (7) (2019) 911.
 - [53] P. Liu, B. Qi, S. Banerjee, Edgeeye: An edge service framework for real-time intelligent video analytics, in: *Proceedings of the 1st International Workshop on Edge Systems, Analytics and Networking*, 2018, pp. 1–6.
 - [54] H. B. Pasandi, T. Nadeem, Convince: Collaborative cross-camera video analytics at the edge, *arXiv preprint arXiv:2002.03797*.
 - [55] H. B. Pasandi, T. Nadeem, Collaborative intelligent cross-camera video analytics at edge: Opportunities and challenges, in: *Proceedings of the First International Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things*, 2019, pp. 15–18.
 - [56] K. Iwanicki, A distributed systems perspective on industrial iot, in: *2018 IEEE 38th International Conference on Distributed Computing Systems (ICDCS)*, IEEE, 2018, pp. 1164–1170.
 - [57] W. Dai, H. Nishi, V. Vyatkin, V. Huang, Y. Shi, X. Guan, Industrial edge computing: Enabling embedded intelligence, *IEEE Industrial Electronics Magazine* 13 (4) (2019) 48–56.
 - [58] T. Qiu, J. Chi, X. Zhou, Z. Ning, M. Atiquzzaman, D. O. Wu, Edge computing in industrial internet of things: Architecture, advances and challenges, *IEEE Communications Surveys & Tutorials* 22 (4) (2020) 2462–2488.
 - [59] J. Wen, L. He, F. Zhu, Swarm robotics control and communications: imminent challenges for next generation smart logistics, *IEEE Communications Magazine* 56 (7) (2018) 102–107.
 - [60] L. Hu, Y. Miao, G. Wu, M. M. Hassan, I. Humar, irobot-factory: An intelligent robot factory based on cognitive manufacturing and edge computing, *Future Generation Computer Systems* 90 (2019) 569–577.
 - [61] Z. Zhao, P. Lin, L. Shen, M. Zhang, G. Q. Huang, Iot edge computing-enabled collaborative tracking system for manufacturing resources in industrial park, *Advanced Engineering Informatics* 43 (2020) 101044.
 - [62] P. Kochovski, V. Stankovski, Supporting smart construction with dependable edge computing infrastructures and applications, *Automation in Construction* 85 (2018) 182–192.
 - [63] E. Bastug, M. Bennis, M. Médard, M. Debbah, Toward interconnected virtual reality: Opportunities, challenges, and enablers, *IEEE Communications Magazine* 55 (6) (2017) 110–117.
 - [64] E. Bertino, S. Banerjee, Artificial intelligence at the edge, *arXiv preprint arXiv:2012.05410*.
 - [65] M. Elawady, A. Sarhan, Mixed reality applications powered by ioe and edge computing: A survey, in: *Internet of Things?Applications and Future*, Springer, 2020, pp. 125–138.
 - [66] J. P. Queralta, J. Raitoharju, T. N. Gia, N. Passalis, T. Westerlund, Autosos: Towards multi-uav systems supporting maritime search and rescue with lightweight ai and edge computing, *arXiv preprint arXiv:2005.03409*.
 - [67] N. Cheng, W. Xu, W. Shi, Y. Zhou, N. Lu, H. Zhou, X. Shen, Air-ground integrated mobile edge networks: Architecture, challenges, and opportunities, *IEEE Communications Magazine* 56 (8) (2018) 26–32.
 - [68] Y. Rizk, M. Awad, E. W. Tunstel, Cooperative heterogeneous multi-robot systems: a survey, *ACM Computing Surveys (CSUR)* 52 (2) (2019) 1–31.
 - [69] Z. Xiong, Y. Zhang, D. Niyato, P. Wang, Z. Han, When mobile blockchain meets edge computing, *IEEE Communications Magazine* 56 (8) (2018) 33–39.
 - [70] A. V. Rivera, A. Refaey, E. Hossain, A blockchain framework for secure task sharing in multi-access edge computing, *IEEE Network*.
 - [71] A. Zhou, Q. Sun, J. Li, Bcedge: Blockchain-based resource management in d2d-assisted mobile edge computing, *Software: Practice and Experience*.
 - [72] Z. Yang, Y. Liu, Y. Chen, Distributed reinforcement learning for noma-enabled mobile edge computing, in: *2020 IEEE International Conference on Communications Workshops (ICC Workshops)*, IEEE, 2020, pp. 1–6.
 - [73] P. K. Sharma, S. Rathore, Y.-S. Jeong, J. H. Park, Softed-

- genet: Sdn based energy-efficient distributed network architecture for edge computing, *IEEE Communications magazine* 56 (12) (2018) 104–111.
- [74] J. Liu, J. Wan, B. Zeng, Q. Wang, H. Song, M. Qiu, A scalable and quick-response software defined vehicular network assisted by mobile edge computing, *IEEE Communications Magazine* 55 (7) (2017) 94–100.
- [75] H. Peng, Q. Ye, X. S. Shen, Sdn-based resource management for autonomous vehicular networks: A multi-access edge computing approach, *IEEE Wireless Communications* 26 (4) (2019) 156–162.
- [76] R. Morabito, V. Cozzolino, A. Y. Ding, N. Beijar, J. Ott, Consolidate iot edge computing with lightweight virtualization, *IEEE Network* 32 (1) (2018) 102–111.
- [77] M. P. Alves, F. C. Delicato, I. L. Santos, P. F. Pires, Lwcoedge: a lightweight virtualization model and collaboration process for edge computing, *World Wide Web* 23 (2) (2020) 1127–1175.
- [78] J. Tang, S. Liu, L. Liu, B. Yu, W. Shi, Lopecs: A low-power edge computing system for real-time autonomous driving services, *IEEE Access* 8 (2020) 30467–30479.
- [79] L. Liu, S. Lu, R. Zhong, B. Wu, Y. Yao, Q. Zhang, W. Shi, Computing systems for autonomous driving: State-of-the-art and challenges, *IEEE Internet of Things Journal*.
- [80] B. Chen, J. Wan, A. Celesti, D. Li, H. Abbas, Q. Zhang, Edge computing in iot-based manufacturing, *IEEE Communications Magazine* 56 (9) (2018) 103–109.
- [81] Y. K. Lee, S.-J. Lee, H. Lee, D. Yoon, Implementation of distributed smart factory platform based on edge computing and opc ua, in: *IECON 2019-45th Annual Conference of the IEEE Industrial Electronics Society*, Vol. 1, IEEE, 2019, pp. 4235–4239.
- [82] S. Yi, Z. Hao, Q. Zhang, Q. Zhang, W. Shi, Q. Li, Lavea: Latency-aware video analytics on edge computing platform, in: *Proceedings of the Second ACM/IEEE Symposium on Edge Computing*, 2017, pp. 1–13.
- [83] X. Zhang, Y. Wang, S. Lu, L. Liu, W. Shi, et al., Openei: An open framework for edge intelligence, in: *2019 IEEE 39th International Conference on Distributed Computing Systems (ICDCS)*, IEEE, 2019, pp. 1840–1851.
- [84] Q. Zhang, X. Zhang, Q. Zhang, W. Shi, H. Zhong, Firework: Big data sharing and processing in collaborative edge environment, in: *2016 Fourth IEEE Workshop on Hot Topics in Web Systems and Technologies (HotWeb)*, IEEE, 2016, pp. 20–25.
- [85] H. Gupta, Z. Xu, U. Ramachandran, Datafog: Towards a holistic data management platform for the iot age at the network edge, in: *{USENIX} Workshop on Hot Topics in Edge Computing (HotEdge 18)*, 2018.
- [86] A. Trivedi, L. Wang, H. Bal, A. Iosup, Sharing and caring of data at the edge, in: *3rd {USENIX} Workshop on Hot Topics in Edge Computing (HotEdge 20)*, 2020.
- [87] Y. Chen, Y. Zhang, Y. Wu, L. Qi, X. Chen, X. Shen, Joint task scheduling and energy management for heterogeneous mobile edge computing with hybrid energy supply, *IEEE Internet of Things Journal* 7 (9) (2020) 8419–8429.
- [88] L. Yang, B. Liu, J. Cao, Y. Sahni, Z. Wang, Joint computation partitioning and resource allocation for latency sensitive applications in mobile edge clouds, *IEEE Transactions on Services Computing*.
- [89] Q. He, G. Cui, X. Zhang, F. Chen, S. Deng, H. Jin, Y. Li, Y. Yang, A game-theoretical approach for user allocation in edge computing environment, *IEEE Transactions on Parallel and Distributed Systems* 31 (3) (2019) 515–529.
- [90] G. Cui, Q. He, X. Xia, P. Lai, F. Chen, T. Gu, Y. Yang, Interference-aware saas user allocation game for edge computing, *IEEE Transactions on Cloud Computing*.
- [91] S. Ranadheera, S. Maghsudi, E. Hossain, Computation offloading and activation of mobile edge computing servers: A minority game, *IEEE Wireless Communications Letters* 7 (5) (2018) 688–691.
- [92] G. Fragkos, S. Lebien, E. E. Tsiropoulou, Artificial intelligent multi-access edge computing servers management, *IEEE Access* 8 (2020) 171292–171304.
- [93] Y. Sun, S. Zhou, Z. Niu, Distributed task replication for vehicular edge computing: Performance analysis and learning-based algorithm, *arXiv preprint arXiv:2002.08833*.
- [94] L. Yang, D. Yang, J. Cao, Y. Sahni, X. Xu, Qos guaranteed resource allocation for live virtual machine migration in edge clouds, *IEEE Access* 8 (2020) 78441–78451.
- [95] J. Yan, N. Li, Z. Zhang, A. X. Liu, J. F. Martinez, X. Yuan, Game theory based joint task offloading and resources allocation algorithm for mobile edge computing, *arXiv preprint arXiv:1912.07599*.
- [96] C. Wang, C. Liang, F. R. Yu, Q. Chen, L. Tang, Computation offloading and resource allocation in wireless cellular networks with mobile edge computing, *IEEE Transactions on Wireless Communications* 16 (8) (2017) 4924–4938.
- [97] X. Lyu, C. Ren, W. Ni, H. Tian, R. P. Liu, X. Tao, Distributed online learning of cooperative caching in edge cloud, *IEEE Transactions on Mobile Computing*.
- [98] W. Jiang, G. Feng, S. Qin, Y.-C. Liang, Learning-based cooperative content caching policy for mobile edge computing, in: *ICC 2019-2019 IEEE International Conference on Communications (ICC)*, IEEE, 2019, pp. 1–6.
- [99] C. Shu, Z. Zhao, G. Min, S. Chen, Mobile edge aided data dissemination for wireless healthcare systems, *IEEE Transactions on Computational Social Systems* 6 (5) (2019) 898–906.
- [100] Q. Yuan, H. Zhou, J. Li, Z. Liu, F. Yang, X. S. Shen, Toward efficient content delivery for automated driving services: An edge computing solution, *IEEE Network* 32 (1) (2018) 80–86.
- [101] L. Yang, L. Zhang, Z. He, J. Cao, W. Wu, Efficient hybrid data dissemination for edge-assisted automated driving, *IEEE Internet of Things Journal* 7 (1) (2019) 148–159.
- [102] A. Aral, T. Ovatman, A decentralized replica placement algorithm for edge computing, *IEEE transactions on network and service management* 15 (2) (2018) 516–529.
- [103] D. Mealha, N. Pregoça, M. C. Gomes, J. Leitão, Data replication on the cloud/edge, in: *Proceedings of the 6th Workshop on Principles and Practice of Consistency for Distributed Data*, 2019, pp. 1–7.
- [104] P. A. Apostolopoulos, E. E. Tsiropoulou, S. Papavassiliou, Cognitive data offloading in mobile edge computing for internet of things, *IEEE Access* 8 (2020) 55736–55749.
- [105] G. Mitsis, E. E. Tsiropoulou, S. Papavassiliou, Data offloading in uav-assisted multi-access edge computing systems: A resource-based pricing and user risk-awareness approach, *Sensors* 20 (8) (2020) 2434.
- [106] M. Zeng, V. Fodor, Dynamic spectrum sharing for load balancing in multi-cell mobile edge computing, *IEEE Wireless Communications Letters* 9 (2) (2019) 189–193.
- [107] M. Liu, F. R. Yu, Y. Teng, V. C. Leung, M. Song, Distributed resource allocation in blockchain-based video streaming systems with mobile edge computing, *IEEE Transactions on Wireless Communications* 18 (1) (2018) 695–708.
- [108] H. Peng, X. Shen, Multi-agent reinforcement learning based resource management in mec-and uav-assisted vehicular networks, *IEEE Journal on Selected Areas in Communications* 39 (1) (2020) 131–141.
- [109] X. Liu, J. Yu, Y. Gao, Multi-agent reinforcement learning for resource allocation in iot networks with edge computing, *arXiv preprint arXiv:2004.02315*.
- [110] Y. Wu, K. Ni, C. Zhang, L. P. Qian, D. H. Tsang, Noma-assisted multi-access mobile edge computing: A joint optimization of computation offloading and time allocation, *IEEE Transactions on Vehicular Technology* 67 (12) (2018) 12244–12258.
- [111] L. P. Qian, B. Shi, Y. Wu, B. Sun, D. H. Tsang, Noma-enabled mobile edge computing for internet of things via joint communication and computation resource allocations, *IEEE Internet of Things Journal* 7 (1) (2019) 718–733.
- [112] L. Qian, Y. Wu, F. Jiang, N. Yu, W. Lu, B. Lin, Noma assisted multi-task multi-access mobile edge computing via deep

- reinforcement learning for industrial internet of things, *IEEE Transactions on Industrial Informatics* 17 (8) (2020) 5688–5698.
- [113] B. Liu, Y. Cao, Y. Zhang, T. Jiang, A distributed framework for task offloading in edge computing networks of arbitrary topology, *IEEE Transactions on Wireless Communications* 19 (4) (2020) 2855–2867.
- [114] X. Lyu, W. Ni, H. Tian, R. P. Liu, X. Wang, G. B. Giannakis, A. Paulraj, Distributed online optimization of fog computing for selfish devices with out-of-date information, *IEEE Transactions on Wireless Communications* 17 (11) (2018) 7704–7717.
- [115] N. Li, J.-F. Martinez-Ortega, V. H. Diaz, Distributed power control for interference-aware multi-user mobile edge computing: a game theory approach, *IEEE Access* 6 (2018) 36105–36114.
- [116] Z. Tan, F. R. Yu, X. Li, H. Ji, V. C. Leung, Virtual resource allocation for heterogeneous services in full duplex-enabled scns with mobile edge computing and caching, *IEEE Transactions on Vehicular Technology* 67 (2) (2017) 1794–1808.
- [117] R. Roostaei, Z. Dabiri, Z. Movahedi, A game-theoretic joint optimal pricing and resource allocation for mobile edge computing in noma-based 5g networks and beyond, *Computer Networks* 198 (2021) 108352.
- [118] X. Pei, H. Yu, X. Wang, Y. Chen, M. Wen, Y.-C. Wu, Noma-based pervasive edge computing: Secure power allocation for iov, *IEEE Transactions on Industrial Informatics* 17 (7) (2020) 5021–5030.
- [119] J. Heydari, V. Ganapathy, M. Shah, Dynamic task offloading in multi-agent mobile edge computing networks, in: 2019 IEEE Global Communications Conference (GLOBECOM), IEEE, 2019, pp. 1–6.
- [120] M. Munir, N. H. Tran, W. Saad, C. S. Hong, et al., Multi-agent meta-reinforcement learning for self-powered and sustainable edge computing systems, *arXiv preprint arXiv:2002.08567*.
- [121] Q. Liu, T. Han, E. Moges, Edgeslice: Slicing wireless edge computing network with decentralized deep reinforcement learning, *arXiv preprint arXiv:2003.12911*.
- [122] Y. Du, J. Li, L. Shi, T. Liu, F. Shu, Z. Han, Two-tier matching game in small cell networks for mobile edge computing, *IEEE Transactions on Services Computing*.
- [123] Z. Yang, Y. Liu, Y. Chen, N. Al-Dhahir, Cache-aided noma mobile edge computing: A reinforcement learning approach, *IEEE Transactions on Wireless Communications* 19 (10) (2020) 6899–6915.
- [124] J. Xu, L. Chen, P. Zhou, Joint service caching and task offloading for mobile edge computing in dense networks, in: IEEE INFOCOM 2018-IEEE Conference on Computer Communications, IEEE, 2018, pp. 207–215.
- [125] X. Yang, Z. Fei, J. Zheng, N. Zhang, A. Anpalagan, Joint multi-user computation offloading and data caching for hybrid mobile cloud/edge computing, *IEEE Transactions on Vehicular Technology* 68 (11) (2019) 11018–11030.
- [126] Y. Zhang, B. Di, Z. Zheng, J. Lin, L. Song, Distributed multi-cloud multi-access edge computing by multi-agent reinforcement learning, *IEEE Transactions on Wireless Communications*.
- [127] X. Huang, L. He, W. Zhang, Vehicle speed aware computing task offloading and resource allocation based on multi-agent reinforcement learning in a vehicular edge computing network, in: 2020 IEEE International Conference on Edge Computing (EDGE), IEEE, 2020, pp. 1–8.
- [128] L. Yang, J. Cao, Z. Wang, W. Wu, Network aware mobile edge computation partitioning in multi-user environments, *IEEE Transactions on Services Computing*.
- [129] Q.-V. Pham, H. T. Nguyen, Z. Han, W.-J. Hwang, Coalitional games for computation offloading in noma-enabled multi-access edge computing, *IEEE Transactions on Vehicular Technology* 69 (2) (2019) 1982–1993.
- [130] P. Lai, Q. He, G. Cui, F. Chen, J. Grundy, M. Abdelrazek, J. G. Hosking, Y. Yang, Cost-effective user allocation in 5g noma-based mobile edge computing systems, *IEEE Transactions on Mobile Computing*.
- [131] X. Xu, J. Liu, X. Tao, Mobile edge computing enhanced adaptive bitrate video delivery with joint cache and radio resource allocation, *IEEE Access* 5 (2017) 16406–16415.
- [132] X. Xia, F. Chen, Q. He, G. Cui, J. C. Grundy, M. Abdelrazek, X. Xu, H. Jin, Data, user and power allocations for caching in multi-access edge computing, *IEEE Transactions on Parallel and Distributed Systems* 33 (5) (2021) 1144–1155.
- [133] A. Ndikumana, N. H. Tran, T. M. Ho, Z. Han, W. Saad, D. Niyato, C. S. Hong, Joint communication, computation, caching, and control in big data multi-access edge computing, *IEEE Transactions on Mobile Computing* 19 (6) (2019) 1359–1374.
- [134] L. N. Huynh, Q.-V. Pham, T. D. Nguyen, M. D. Hossain, Y.-R. Shin, E.-N. Huh, Joint computational offloading and data-content caching in noma-mec networks, *IEEE Access* 9 (2021) 12943–12954.
- [135] L. Buşoni, R. Babuška, B. De Schutter, Multi-agent reinforcement learning: An overview, *Innovations in multi-agent systems and applications-1* (2010) 183–221.
- [136] N. Naderializadeh, M. Hashemi, Energy-aware multi-server mobile edge computing: A deep reinforcement learning approach, in: 2019 53rd Asilomar Conference on Signals, Systems, and Computers, IEEE, 2019, pp. 383–387.
- [137] Y. Wang, H. Liu, W. Zheng, Y. Xia, Y. Li, P. Chen, K. Guo, H. Xie, Multi-objective workflow scheduling with deep-q-network-based multi-agent reinforcement learning, *IEEE Access* 7 (2019) 39974–39982.
- [138] Z. Chen, X. Wang, Decentralized computation offloading for multi-user mobile edge computing: A deep reinforcement learning approach, *EURASIP Journal on Wireless Communications and Networking* 2020 (1) (2020) 1–21.
- [139] H. Gao, X. Wang, X. Ma, W. Wei, S. Mumtaz, Com-ddpg: A multiagent reinforcement learning-based offloading strategy for mobile edge computing, *arXiv preprint arXiv:2012.05105*.
- [140] H. Lu, C. Gu, F. Luo, W. Ding, S. Zheng, Y. Shen, Optimization of task offloading strategy for mobile edge computing based on multi-agent deep reinforcement learning, *IEEE Access* 8 (2020) 202573–202584.
- [141] Z. Cao, P. Zhou, R. Li, S. Huang, D. Wu, Multiagent deep reinforcement learning for joint multichannel access and task offloading of mobile-edge computing in industry 4.0, *IEEE Internet of Things Journal* 7 (7) (2020) 6201–6213.
- [142] X. Wang, Z. Ning, S. Guo, Multi-agent imitation learning for pervasive edge computing: a decentralized computation offloading algorithm, *IEEE Transactions on Parallel and Distributed Systems* 32 (2) (2020) 411–425.
- [143] X. Wang, C. Wang, X. Li, V. C. Leung, T. Taleb, Federated deep reinforcement learning for internet of things with decentralized cooperative edge caching, *IEEE Internet of Things Journal*.
- [144] R. Wang, M. Li, L. Peng, Y. Hu, M. M. Hassan, A. Alelaiwi, Cognitive multi-agent empowering mobile edge computing for resource caching and collaboration, *Future Generation Computer Systems* 102 (2020) 66–74.
- [145] P. Dai, K. Liu, X. Wu, H. Xing, Z. Yu, V. C. Lee, A learning algorithm for real-time service in vehicular networks with mobile-edge computing, in: ICC 2019-2019 IEEE International Conference on Communications (ICC), IEEE, 2019, pp. 1–6.
- [146] X. Zhu, Y. Luo, A. Liu, M. Z. A. Bhuiyan, S. Zhang, Multi-agent deep reinforcement learning for vehicular computation offloading in iot, *IEEE Internet of Things Journal*.
- [147] M. Munir, S. F. Abedin, N. H. Tran, Z. Han, E. N. Huh, C. S. Hong, et al., Risk-aware energy scheduling for edge computing with microgrid: A multi-agent deep reinforcement learning approach, *arXiv preprint arXiv:2003.02157*.
- [148] S. Ranadheera, S. Maghsudi, E. Hossain, Mobile edge computation offloading using game theory and reinforcement learning, *arXiv preprint arXiv:1711.09012*.
- [149] M. Hu, Z. Xie, D. Wu, Y. Zhou, X. Chen, L. Xiao, Heterogeneous edge offloading with incomplete information: A minor-

- ity game approach, *IEEE Transactions on Parallel and Distributed Systems* 31 (9) (2020) 2139–2154.
- [150] E. Abdelhalim, M. Obayya, S. Kishk, Distributed fog-to-cloud computing system: A minority game approach, *Concurrency and Computation: Practice and Experience* 31 (15) (2019) e5162.
- [151] P. A. Apostolopoulos, E. E. Tsiropoulou, S. Papavassiliou, Game-theoretic learning-based qos satisfaction in autonomous mobile edge computing, in: 2018 Global Information Infrastructure and Networking Symposium (GIIS), IEEE, 2018, pp. 1–5.
- [152] H. Cao, J. Cai, Distributed multiuser computation offloading for cloudlet-based mobile cloud computing: A game-theoretic machine learning approach, *IEEE Transactions on Vehicular Technology* 67 (1) (2017) 752–764.
- [153] L. Yang, H. Zhang, X. Li, H. Ji, V. C. Leung, A distributed computation offloading strategy in small-cell networks integrated with mobile edge computing, *IEEE/ACM Transactions on Networking* 26 (6) (2018) 2762–2773.
- [154] M. Deng, H. Tian, X. Lyu, Adaptive sequential offloading game for multi-cell mobile edge computing, in: 2016 23rd International Conference on Telecommunications (ICT), IEEE, 2016, pp. 1–5.
- [155] S. Zhou, W. Jadoon, The partial computation offloading strategy based on game theory for multi-user in mobile edge computing environment, *Computer Networks* 178 (2020) 107334.
- [156] T. Ouyang, Z. Zhou, X. Chen, Follow me at the edge: Mobility-aware dynamic service placement for mobile edge computing, *IEEE Journal on Selected Areas in Communications* 36 (10) (2018) 2333–2345.
- [157] M. Liu, Y. Liu, Price-based distributed offloading for mobile-edge computing with computation capacity constraints, *IEEE Wireless Communications Letters* 7 (3) (2017) 420–423.
- [158] X. Chen, L. Jiao, W. Li, X. Fu, Efficient multi-user computation offloading for mobile-edge cloud computing, *IEEE/ACM Transactions on Networking* 24 (5) (2015) 2795–2808.
- [159] Q. Xu, Z. Su, Q. Zheng, M. Luo, B. Dong, K. Zhang, Game theoretical secure caching scheme in multihoming edge computing-enabled heterogeneous networks, *IEEE Internet of Things Journal* 6 (3) (2018) 4536–4546.
- [160] Y. Jiang, Y. Hu, M. Bennis, F.-C. Zheng, X. You, A mean field game-based distributed edge caching in fog radio access networks, *IEEE Transactions on Communications* 68 (3) (2019) 1567–1580.
- [161] Y. Zhan, S. Guo, P. Li, J. Zhang, A deep reinforcement learning based offloading game in edge computing, *IEEE Transactions on Computers* 69 (6) (2020) 883–893.
- [162] D. Shi, H. Gao, L. Wang, M. Pan, Z. Han, H. V. Poor, Mean field game guided deep reinforcement learning for task placement in cooperative multiaccess edge computing, *IEEE Internet of Things Journal* 7 (10) (2020) 9330–9340.
- [163] A. Asheralieva, D. Niyato, Hierarchical game-theoretic and reinforcement learning framework for computational offloading in uav-enabled mobile edge computing networks with multiple service providers, *IEEE Internet of Things Journal* 6 (5) (2019) 8753–8769.
- [164] M.-A. Messous, H. Sedjelmaci, N. Houari, S.-M. Senouci, Computation offloading game for an uav network in mobile edge computing, in: 2017 IEEE International Conference on Communications (ICC), IEEE, 2017, pp. 1–6.
- [165] Y. Wang, P. Lang, D. Tian, J. Zhou, X. Duan, Y. Cao, D. Zhao, A game-based computation offloading method in vehicular multiaccess edge computing networks, *IEEE Internet of Things Journal* 7 (6) (2020) 4987–4996.
- [166] Y. Liu, S. Wang, J. Huang, F. Yang, A computation offloading algorithm based on game theory for vehicular edge networks, in: 2018 IEEE International Conference on Communications (ICC), IEEE, 2018, pp. 1–6.
- [167] W. Chen, Z. Zhang, Z. Hong, C. Chen, J. Wu, S. Maharjan, Z. Zheng, Y. Zhang, Cooperative and distributed computation offloading for blockchain-empowered industrial internet of things, *IEEE Internet of Things Journal* 6 (5) (2019) 8433–8446.
- [168] D. Zhang, L. Tan, J. Ren, M. K. Awad, S. Zhang, Y. Zhang, P.-J. Wan, Near-optimal and truthful online auction for computation offloading in green edge-computing systems, *IEEE Transactions on Mobile Computing* 19 (4) (2019) 880–893.
- [169] L. Lei, H. Xu, X. Xiong, K. Zheng, W. Xiang, X. Wang, Multiuser resource control with deep reinforcement learning in iot edge computing, *IEEE Internet of Things Journal* 6 (6) (2019) 10119–10133.
- [170] P. Kayal, J. Liebeherr, Distributed service placement in fog computing: An iterative combinatorial auction approach, in: 2019 IEEE 39th International Conference on Distributed Computing Systems (ICDCS), IEEE, 2019, pp. 2145–2156.
- [171] Y. Xiao, M. Krunz, Distributed optimization for energy-efficient fog computing in the tactile internet, *IEEE Journal on Selected Areas in Communications* 36 (11) (2018) 2390–2400.
- [172] P. Kayal, J. Liebeherr, Autonomic service placement in fog computing, in: 2019 IEEE 20th International Symposium on "A World of Wireless, Mobile and Multimedia Networks" (WoWMoM), IEEE, 2019, pp. 1–9.
- [173] Q. Cui, J. Zhang, X. Zhang, K.-C. Chen, X. Tao, P. Zhang, Online anticipatory proactive network association in mobile edge computing for iot, *IEEE Transactions on Wireless Communications* 19 (7) (2020) 4519–4534.
- [174] X. Lyu, C. Ren, W. Ni, H. Tian, R. P. Liu, Distributed optimization of collaborative regions in large-scale inhomogeneous fog computing, *IEEE Journal on Selected Areas in Communications* 36 (3) (2018) 574–586.
- [175] C. Ren, X. Lyu, W. Ni, H. Tian, W. Song, R. P. Liu, Distributed online optimization of fog computing for internet of things under finite device buffers, *IEEE Internet of Things Journal* 7 (6) (2020) 5434–5448.
- [176] C. Ren, X. Lyu, W. Ni, H. Tian, R. P. Liu, Profitable cooperative region for distributed online edge caching, *IEEE Transactions on Communications* 67 (7) (2019) 4696–4708.
- [177] S. Sardellitti, G. Scutari, S. Barbarossa, Joint optimization of radio and computational resources for multicell mobile-edge computing, *IEEE Transactions on Signal and Information Processing over Networks* 1 (2) (2015) 89–103.
- [178] M. Li, N. Cheng, J. Gao, Y. Wang, L. Zhao, X. Shen, Energy-efficient uav-assisted mobile edge computing: Resource allocation and trajectory optimization, *IEEE Transactions on Vehicular Technology* 69 (3) (2020) 3424–3438.
- [179] P. Leitão, J. Barbosa, D. Trentesaux, Bio-inspired multi-agent systems for reconfigurable manufacturing systems, *Engineering Applications of Artificial Intelligence* 25 (5) (2012) 934–944.
- [180] D. J. Dubois, G. Valetto, D. Lucia, E. Di Nitto, Myocloud: Elasticity through self-organized service placement in decentralized clouds, in: 2015 IEEE 8th International Conference on Cloud Computing, IEEE, 2015, pp. 629–636.
- [181] T. Welsh, E. Benkhelifa, Bioinspired multiagent embryonic architecture for resilient edge networks, *IEEE Transactions on Industrial Informatics* 15 (12) (2019) 6344–6353.
- [182] J. Li, H. Zhao, A. S. Hafid, J. Wei, H. Yin, B. Ren, A bio-inspired solution to cluster-based distributed spectrum allocation in high-density cognitive internet of things, *IEEE Internet of Things Journal* 6 (6) (2019) 9294–9307.
- [183] P. Di Lorenzo, S. Barbarossa, A bio-inspired swarming algorithm for decentralized access in cognitive radio, *IEEE Transactions on Signal Processing* 59 (12) (2011) 6160–6174.
- [184] N. Jabeur, A. U.-H. Yasar, E. Shakshuki, H. Haddad, Toward a bio-inspired adaptive spatial clustering approach for iot applications, *Future Generation Computer Systems* 107 (2020) 736–744.
- [185] C.-J. Lee, J.-Y. Jung, J.-R. Lee, Bio-inspired distributed transmission power control considering qos fairness in wireless body area sensor networks, *Sensors* 17 (10) (2017) 2344.
- [186] Y. Zhang, W. Xia, F. Yan, H. Cheng, L. Shen, Multi-agent reinforcement learning for joint wireless and computational re-

- source allocation in mobile edge computing system, in: International Conference on Ad Hoc Networks, Springer, 2019, pp. 149–161.
- [187] M. G. R. Alam, Y. K. Tun, C. S. Hong, Multi-agent and reinforcement learning based code offloading in mobile fog, in: 2016 International Conference on Information Networking (ICOIN), IEEE, 2016, pp. 285–290.
- [188] J. Cui, Y. Liu, A. Nallanathan, Multi-agent reinforcement learning-based resource allocation for uav networks, IEEE Transactions on Wireless Communications 19 (2) (2019) 729–743.
- [189] D. A. Temesgene, M. Miozzo, D. Gunduz, P. Dini, Distributed deep reinforcement learning for functional split control in energy harvesting virtualized small cells, IEEE Transactions on Sustainable Computing.
- [190] Y. Su, W. Fan, Y. Liu, F. Wu, Game-based distributed pricing and task offloading in multi-cloud and multi-edge environments, Computer Networks 200 (2021) 108523.
- [191] Z. Cao, H. Zhang, B. Liu, B. Sheng, Revenue sharing in edge-cloud systems: A game-theoretic perspective, Computer Networks 176 (2020) 107286.
- [192] M. Guo, Q. Li, Z. Peng, X. Liu, D. Cui, Energy harvesting computation offloading game towards minimizing delay for mobile edge computing, Computer Networks 204 (2022) 108678.
- [193] B. Xiang, J. Elias, F. Martignon, E. Di Nitto, Resource calendaring for mobile edge computing: Centralized and decentralized optimization approaches, Computer Networks 199 (2021) 108426.
- [194] T. Rausch, S. Nastic, S. Dustdar, Emma: Distributed qos-aware mqtt middleware for edge computing applications, in: 2018 IEEE International Conference on Cloud Engineering (IC2E), IEEE, 2018, pp. 191–197.
- [195] A. Mukherjee, N. Dey, D. De, Edgedrone: Qos aware mqtt middleware for mobile edge computing in opportunistic internet of drone things, Computer Communications 152 (2020) 93–108.
- [196] B. P. Rimal, M. Maier, M. Satyanarayanan, Experimental testbed for edge computing in fiber-wireless broadband access networks, IEEE Communications Magazine 56 (8) (2018) 160–167.
- [197] T. Hao, Y. Huang, X. Wen, W. Gao, F. Zhang, C. Zheng, L. Wang, H. Ye, K. Hwang, Z. Ren, et al., Edge aibench: towards comprehensive end-to-end edge computing benchmarking, in: International Symposium on Benchmarking, Measuring and Optimization, Springer, 2018, pp. 23–30.
- [198] L. T. Bolivar, C. Tselios, D. M. Area, G. Tsolis, On the deployment of an open-source, 5g-aware evaluation testbed, in: 2018 6th IEEE International Conference on Mobile Cloud Computing, Services, and Engineering (MobileCloud), IEEE, 2018, pp. 51–58.
- [199] H. Gupta, A. Vahid Dastjerdi, S. K. Ghosh, R. Buyya, ifogsim: A toolkit for modeling and simulation of resource management techniques in the internet of things, edge and fog computing environments, Software: Practice and Experience 47 (9) (2017) 1275–1296.
- [200] C. Sonmez, A. Ozgovde, C. Ersoy, Edgecloudsim: An environment for performance evaluation of edge computing systems, Transactions on Emerging Telecommunications Technologies 29 (11) (2018) e3493.
- [201] T. Qayyum, A. W. Malik, M. A. K. Khattak, O. Khalid, S. U. Khan, Fognetsim++: A toolkit for modeling and simulation of distributed fog environment, IEEE Access 6 (2018) 63570–63583.
- [202] Z. Xiong, S. Feng, D. Niyato, P. Wang, Z. Han, Optimal pricing-based edge computing resource management in mobile blockchain, in: 2018 IEEE International Conference on Communications (ICC), IEEE, 2018, pp. 1–6.
- [203] M. Chen, Z. Yang, W. Saad, C. Yin, H. V. Poor, S. Cui, A joint learning and communications framework for federated learning over wireless networks, IEEE Transactions on Wireless Communications.
- [204] Y. Zhan, P. Li, Z. Qu, D. Zeng, S. Guo, A learning-based incentive mechanism for federated learning, IEEE Internet of Things Journal 7 (7) (2020) 6360–6368.